RetailPulse: Agentic AI for Retail Store Intelligence

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Students' Submission

We certify that this report is submitted as the FYP-1 mid-report for the project titled *RetailPulse: Agentic AI for Retail Store Intelligence*.

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Anti-Plagiarism Statement

The students certify that this report is their own work and that all sources used in preparing the report have been appropriately acknowledged and referenced.

Authors' Declaration

We declare that the work documented in this report represents the work completed by the project team up to the FYP-1 mid-report stage. Contributions, tools, external components, and relevant sources are disclosed where applicable.

Abstract

Most retail CCTV footage is used only for theft prevention, and small and mid-sized retailers cannot afford specialised analytics hardware or analysts to find out whether customers are being attended to. This report presents *RetailPulse*, a CPU-only system that aims to turn existing CCTV footage into customer-engagement, staff-performance and store-traffic analytics without new cameras. The design separates detection, classification, engagement and trajectory analysis into vision components behind a Next.js dashboard for managers and employees, with FastAPI and SQLite for application records.

The project follows an iterative approach. At the FYP-1 mid-report stage the web application with role-aware access and an interaction pipeline are implemented. The pipeline combines a trained employee/customer YOLO detector (mAP50 0.969 on 455 held-out images), BoT-SORT tracking, body-height proximity with a pass-by filter, and a YOLO pose judge that confirms an interaction only when someone faces the other person. On the one labelled clip the judge raises interaction precision from 0.50 to 1.00; the pipeline ran on all 11 local clips and 44 unit tests pass. A floor-distance prototype was built and then descoped because the moving-camera footage supported almost no floor projections. Dashboard vision metrics still use prototype data. FYP-2 will integrate the vision agents with the dashboard, evaluate them on labelled clips from the supplied videos, and harden the system for the final demonstration.

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Executive Summary

Problem and motivation. Retail stores record CCTV footage mainly for security, yet managers lack a low-cost way to know whether customers are attended to, how staff time is distributed, and which zones receive traffic. Commercial retail-analytics products typically depend on dedicated sensors or hardware and on subscription costs, which puts them out of reach of many small and mid-sized retailers.

Proposed solution. RetailPulse reuses existing CCTV infrastructure. Vision components for Detection, Classification, Engagement and Trajectory convert video into per-entity analytics: footfall, approach time, interaction duration, idle time, unattended-customer flags, dwell time, paths, and heatmaps. Results are presented on a web dashboard for managers and a focused workspace for employees.

Scope. The committed scope is a single-store prototype with three cameras, running on CPU only. Multi-store scaling, full cross-camera re-identification, and shelf-level interaction detection are deliberately deferred to future work after a feasibility triage against hardware, timeline, and dependency constraints.

Current status. The Next.js frontend covers the manager, employee, and shared pages. FastAPI and SQLite support account, camera, alert and operational-task workflows. Role-aware access is enforced with signed HttpOnly session cookies and an internal secret header. A local interaction pipeline combines a trained employee/customer detector, BoT-SORT tracking, body-height proximity with a pass-by filter, and a YOLO pose judge that confirms an interaction only when someone faces the other person. A metric floor-distance prototype was built and then descoped because the moving-camera footage supported almost no floor projections. The dashboard vision metrics still partly use prototype or demo data.

Validation, risks, and remaining work. The detector reaches mAP50 0.969 on 455 held-out images. On the one labelled clip the pose judge raises interaction precision from 0.50 to 1.00, and the pipeline completes all 11 local clips; 44 interaction unit tests pass. Accuracy is measured on one labelled clip only. The main risks are CPU-only inference speed, role-classification accuracy, depth ambiguity in a single image, and identity switches on moving footage. The remaining work covers vision-agent integration, evaluation on store footage, hardening, and the final demonstration.

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List of Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
CCTV	Closed-Circuit Television
CLI	Command-Line Interface
CPU	Central Processing Unit
CV	Computer Vision
DEI	Diversity, Equity and Inclusion
FR	Functional Requirement
FYP	Final Year Project
GPU	Graphics Processing Unit
HSV	Hue, Saturation, Value (colour space)
JSON	JavaScript Object Notation
JSONL	JSON Lines (one JSON record per line)
KPI	Key Performance Indicator
mAP	Mean Average Precision
MAE	Mean Absolute Error
MVP	Minimum Viable Product
ORB	Oriented FAST and Rotated BRIEF
RANSAC	Random Sample Consensus
NFR	Non-Functional Requirement
PDF	Portable Document Format
Re-ID	Re-Identification
UI	User Interface
YOLO	You Only Look Once (object detector family)

Chapter 1

Introduction and Project Vision

1.1 Project Overview

RetailPulse (*Agentic AI for Retail Store Intelligence*) is a system that converts existing in-store CCTV footage into live customer-engagement analytics, staff-performance insight, and store-traffic intelligence, without new cameras or specialised hardware. It is a Final Year Project that explores practical use of multi-agent AI systems in a resource-constrained retail environment.

The system assumes a three-camera setup: one entrance-facing camera and two floor-facing cameras. Their feeds pass through a pipeline of specialised agents. The mid-iteration work covers person/role detection, interaction detection (proximity plus a pose-based facing check) and the web interfaces for managers and employees.

1.2 Problem Statement

Retail store owners and managers have no low-cost way to know whether customers on the floor are being attended to, how staff time is split between active engagement and idle time, and which areas of the store attract traffic and which are ignored. Existing CCTV footage is recorded but rarely reviewed, and dedicated retail-analytics hardware or analysts are often financially out of reach for small and mid-sized retailers.

1.3 Problem Context and Motivation

CCTV systems in most stores exist for theft prevention. The footage holds operational information (queues, approaches, dwell, idle time) that goes unused. Managers who want

this information must watch footage manually, which does not scale, or buy specialised products.

The motivation is threefold. First, service quality: customers who are not approached in a reasonable time may leave without buying. Second, fair and informed staffing: managers need visibility into floor coverage and idle periods. Third, accessibility: a CPU-only solution built on existing cameras lowers the barrier for retailers that cannot invest in new capital equipment. The project also emphasises privacy-conscious design, using non-biometric classification and presenting results as decision support rather than automated employee evaluation. Service expectations vary by store: frequent approaches may be inappropriate in luxury or low-contact settings. Unattended-customer flags and staff scores must therefore be interpreted against a store-defined service policy, not a universal expectation that staff should constantly approach shoppers.

1.4 Project Vision

RetailPulse aims to make store intelligence accessible to retailers of any size by reusing the cameras they already own, running on commodity CPUs, and giving managers clear and trustworthy answers to everyday operational questions.

The project uses two distinct YOLO model roles: a YOLO pose model supplies body-keypoint evidence for interaction analysis, while a standard YOLO object-detection model with customer/employee classes supplies person boxes and role labels. The latter is a detector trained on role classes, not a separate image-classification-only network. Interaction events are derived from pose/proximity and temporal rules rather than treating a pose prediction itself as an interaction label.

1.5 Project Objectives

The project has the following measurable objectives.

O1. Count footfall from the entrance camera and report it per time interval.

O2. Detect people in each processed frame and classify them as staff or customer without biometric identification; the interaction pipeline uses a trained employee/customer detector, while earlier demo paths use zone and appearance heuristics.

O3. Compute engagement metrics: approach time, interaction duration, idle time, unattended-customer flags, and an employee score.

O4. Track movement to produce paths, zone dwell times, heatmaps, and crowd density.

O5. Provide a manager dashboard, an employee workspace, alerts, and exportable reports with narrative summaries.

O6. Detect employee–customer interactions by combining body-height-normalised proximity, duration and a pose-based facing check, and reject close pass-bys.

O7. Run the vision pipeline on CPU only.

1.6 Project Scope

In scope:

- Footfall counting (entrance camera).
- Staff/customer classification (non-biometric, including the trained role detector used by the interaction pipeline).
- Engagement analytics: approach time, interaction duration, idle detection, unattended flagging, and employee scoring.
- Trajectory and spatial behaviour: path tracking, zone dwell time, heatmaps, and crowd density.
- Interaction detection from body-height-normalised proximity, duration, a pass-by filter and a pose-based facing check, run through local command-line tools.
- Web application with manager and employee roles.

Explicitly out of scope (see Chapter 9 and Future Work):

- Cross-camera identity association; this is reserved for the next phase.
- Metric floor-plane distance estimation; a calibrated-homography prototype was built and then descoped (Section 6.4.3).
- Full three-dimensional reconstruction and per-pixel depth-map generation.
- Shelf-level product interaction detection.
- Long-term store layout and promotional-effectiveness evaluation.
- Multi-store scaling and real-time processing as the primary deliverable (the current pipeline is offline/batch).

These exclusions were chosen after a feasibility triage against hardware (no dedicated GPU), timeline, and dependency constraints.

1.7 Project Modules

Table 1.1: Major modules and responsibilities

Module	Responsibility
Detection Agent	Person detection in each frame using a YOLO-based detector.
Classification Agent	Non-biometric staff/customer classification; the interaction pipeline uses the trained role detector, and earlier demo paths use heuristics.
Engagement Agent	Interaction duration, approach time, and idle detection.
Trajectory Agent	Movement paths, zone dwell time, and heatmap generation.
Insight Narration Agent	Generates natural-language report summaries on demand.
Interaction pipeline	Role-labelled tracking, body-height proximity, pass-by filter and YOLO pose judge; local command-line tool with video and JSON outputs.
Web Application	Next.js manager dashboard, employee workspace, alerts, reports, and settings.

1.8 Stakeholders and User Classes

Table 1.2: Stakeholders and user classes

Stakeholder	Characteristics and needs
Store manager	Primary user. Needs store-wide analytics, alerts, and reports. Not necessarily technical.
Employee	Needs a personal view of shift, score, trend, and zone assignment. Sees only employee-relevant data.
Store owner	Interested in service quality, staffing, and floor-coverage decisions at a summary level.
Customers	Appear in footage. Are not identified biometrically. Affected indirectly through service quality and privacy.
Project team and supervisor	Develop, evaluate, and supervise the system.

1.9 Success Criteria

The supervisor agreed to the 70-video dataset shown by the team. No numeric classification/interaction accuracy or dashboard response-time threshold was agreed. The criteria below describe the evaluation evidence to report; dataset acceptance does not establish model accuracy.

- The pipeline processes the supplied videos end to end on a CPU-only machine.
- Staff/customer classification and interaction detection are evaluated against manually labelled clips, with accuracy reported. No numeric pass threshold has been agreed; report the measured results and evaluation method without inventing a target.
- Interaction detection is benchmarked with and without the pose judge on labelled clips, reporting precision, recall and the effect of the judge.
- Dashboard metrics should correspond to the relevant tracked entities and source observations.
- Manager-only routes are not reachable without a valid signed session.
- Reports can be generated and exported as PDF.

1.10 Constraints and Assumptions

Constraints. No dedicated GPU is available. The timeline is limited to the two FYP semesters. The available video dataset constrains evaluation coverage. Processing is offline/batch rather than real time.

Assumptions. Cameras are fixed, and the store has an entrance-facing camera and two floor-facing cameras. Camera views overlap or adjoin so that people leaving one view enter another within a bounded time window. Staff are distinguishable by zone and appearance cues (for example, working areas and uniforms).

Distance assumption. Interaction distance is measured in image space and normalised by body height, so it is dimensionless and cannot fully resolve depth. A metric floor-distance prototype, which assumed an approximately planar floor and a fixed camera, was descoped (Section 6.4.3) because the moving-camera footage supported almost no floor projections.

1.11 Ethical, Legal and DEI Considerations

Privacy. The system does not use face recognition or any other biometric identification. The trained role detector uses learned appearance-based role labels without facial identity storage; earlier demo classification paths use zone/appearance heuristics. Runtime footage remains local and is excluded from Git.

Fair use of results. The system is designed for manager decision support rather than automated employee evaluation. Employee scores are indicators for discussion, and the limitations of the heuristics must be communicated to users.

Legal and consent. Deployment requires a review of applicable law, store policy and appropriate notice/consent arrangements. Pakistan's Ministry of Information Technology and Telecommunication lists personal-data-protection material as draft legislation on its official legislation page [1]. That listing is not a comprehensive statement of all applicable law. The prototype follows data-minimisation principles informed by the GDPR [2]; this design reference does not establish legal compliance or a lawful basis for deployment.

Accessibility and inclusion. The web interface uses readable fonts, clear contrast, and charts with textual values. Appearance-based classification can perform unevenly across clothing styles, and this is recorded as a fairness-related limitation to evaluate in FYP-2.

Chapter 2

Research and Existing Solutions

2.1 Research Method / Approach

Research combined three activities: (i) reviewing the foundational literature on object detection, multi-object tracking, person re-identification, pose-based orientation cues, and planar visual metrology (the last for the descoped floor prototype); (ii) surveying publicly described retail-analytics products and their stated capabilities; and (iii) a feasibility triage of candidate features against hardware (no dedicated GPU), timeline, and dependency constraints. Tools and libraries were then selected by prototyping on the supplied store videos.

2.2 Existing Solutions / Products

Commercial retail analytics. RetailNext describes traffic counts, staffing and interaction-rate analytics [3]. Sensormatic describes shopper traffic and journey insights, including labour and store-traffic information [4]. Amazon Just Walk Out describes camera/sensor-based shopping and automated checkout, including weight-sensor and RFID variants [5]. These are vendor descriptions, not independently reproduced performance results.

Research building blocks. The YOLO family [6] provides fast single-stage detectors with CPU-feasible small variants, available through the Ultralytics implementation [7]. SORT [8] and ByteTrack [9] associate detections across frames into tracks. Person re-identification across cameras is surveyed in [10] and typically depends on learned embeddings and substantial compute.

2.3 Comparison of Existing Solutions

Table 2.1: Qualitative comparison using official product descriptions, checked 20 October 2026

Approach	Documented focus	Evidence boundary
RetailNext	Traffic, conversion, staffing and interaction-rate analytics [3]	Pricing and CCTV compatibility not assessed.
Sensormatic	Shopper traffic/journeys and staffing context [4]	No claim of absent engagement features.
Just Walk Out	Shopper/product sensing and automated checkout [5]	Different checkout-oriented objective; cost not assessed.
RetailPulse prototype	Local-video role tracking and pose-checked interaction detection	CPU smoke tests and local UI evidence; no commercial accuracy or cost comparison.

2.4 Limitations and Gaps

The reviewed commercial products already provide substantial retail analytics, including interaction-related metrics in RetailNext. The project's design objective is therefore an inspectable academic prototype using local footage and explicit heuristic rules, rather than a claim that commercial engagement analytics do not exist. Public product pages alone do not establish relative cost, deployment effort, privacy compliance or comparative accuracy.

The technical gap the project isolates is narrower: separating a service interaction from a close pass-by using only a single ordinary camera. Proximity alone nominates any employee and customer who stay near each other, including two people walking away side by side; on the one labelled clip it reached precision 0.50 (Table 7.5). The project adds a pose-based facing check and measures its effect with and without it, so the contribution is an inspectable, locally run pipeline with a stated evaluation method rather than a new detector.

2.5 Proposed Solution

RetailPulse separates detection, classification, engagement, trajectory and interaction judging into distinct components. The mid-iteration implementation concentrates on these person-analysis components and the web interfaces. Lightweight YOLO detection with local tracker identifiers (BoT-SORT in the interaction pipeline, ByteTrack in the descoped floor prototype) keeps the pipeline CPU-friendly.

2.6 Complex Computing Problem

The complex computing problem is to derive reliable, per-person service and movement analytics from low-cost, uncontrolled CCTV video under tight compute limits. The problem involves several interacting difficulties:

- Conflicting requirements: accuracy against CPU-only throughput, and informativeness against privacy (no biometrics).
- Ambiguous ground truth: whether a staff member near a customer constitutes an "interaction" must be defined operationally from distance, orientation, and duration.
- Consistency of observations: role labels, boxes, distance estimates and interaction cues must refer to the same tracked person despite occlusions and identity switches.
- Non-technical users: results must be presented in a way that managers can act on without over-trusting heuristics.

2.7 Technical Challenges

- **CPU-only inference:** frame sampling and small detector variants are needed to keep batch processing practical.
- **Staff/customer separation without biometrics:** both learned appearance-based role detection and earlier zone rules require evaluation against clothing similarity and staff outside their expected zones.
- **Perspective and depth ambiguity:** pixel separations depend on image location and people at different depths can look close, so the pipeline uses body-height-normalised distance; a calibrated floor mapping to metres was tried and descoped.

- **Floor-contact uncertainty:** seated people, hidden feet, chairs, overlap and frame truncation made box-bottom estimates unreliable in the floor prototype and still affect the body bands used for proximity.
- **Camera movement:** a reference homography became misaligned on the moving-camera footage and insufficient static floor features meant unverified alignment, which led to descoping the floor prototype. Camera movement also causes identity switches.
- **Identity switches and occlusion:** ID switching in crowded scenes affects interaction durations and dwell times.
- **Security of the web tier:** manager-only operational data must not be unlocked by client-side state alone.

2.8 Research Findings and Design Decisions

2.8.1 Interaction Detection Findings

Proximity alone cannot separate a service interaction from a close pass-by. On the labelled clip IMG_8829 it produced one true and one false interaction (precision 0.50). Adding the YOLO pose model as a judge, which requires a head to point within 60° of the other person's head in at least half of four crops, removed the false positive (precision 1.00). This single-clip result motivates the design but does not establish its accuracy; FYP-2 evaluates it on more labelled clips (Chapter 7).

2.8.2 Geometric Basis of the Descoped Floor-Distance Prototype

The floor-distance prototype was descoped (Section 6.4.3); its research basis is kept because it explains why body-height-normalised distance replaced it.

Projective geometry provides a mapping between two views of a plane through a homography [11, 12]. Single-view metrology establishes broader foundations for recovering geometric measurements from perspective images using scene information [13]; the implemented prototype specifically uses explicit image/floor landmark correspondences rather than reproducing that paper's complete procedure. RetailPulse selects planar floor mapping because the intended quantity is separation along the floor and the existing camera and detector can be reused. It does not estimate a depth map or reconstruct arbitrary 3D scene geometry.

Robust fitting with RANSAC [14] supports additional calibration correspondences and reference-alignment checks. ByteTrack [9] supplies per-camera association, while ORB [15] supplies image features for the conservative motion monitor. These are established building blocks; the project's contribution is their integration with explicit position-quality states, configurable calibration and diagnostic exports. Published benchmark results for these methods are not claimed as RetailPulse performance.

Table 2.2: Findings and the decisions they influenced

Finding	Decision
YOLO detectors have small variants usable without a GPU [6, 7].	Use a lightweight YOLO-based detector and sample frames.
ByteTrack [9] gives stable local IDs without a separate appearance network.	Use ByteTrack for per-camera tracking.
Learned re-identification [10] is compute-heavy and biometric-adjacent.	Keep cross-camera association outside the mid-iteration deliverables.
Mixed vision responsibilities are harder to inspect and test.	Separate detection, role classification, interaction and distance logic.
Appearance evidence can be ambiguous.	Keep role confidence and source evidence visible for review.
Client-side role flags are easily altered.	Issue server-signed HttpOnly session cookies and require an internal secret on direct backend calls.
Managers need explanations, not only numbers.	Add a narration agent over camera-derived operational evidence.
Perspective changes pixel scale across the image [11, 12].	Projected foot proxies into floor metres in the prototype; descoped and replaced by body-height-normalised distance.
Proximity alone nominates close pass-bys as interactions.	Add a pass-by filter and a pose judge, and benchmark with and without the judge.
Supplied floor dimensions determine metric scale.	Stored editable landmarks and dimensions in the descoped prototype, with estimated inputs identified in configuration.
Camera movement can invalidate the reference mapping.	Checked alignment with ORB/RANSAC in the prototype and suppressed output when shifted or unverified; almost all frames were rejected, so the prototype was descoped.

Chapter 3

Proposal Feedback and Iteration Plan

3.1 Proposal Feedback Summary

At the proposal defence on 11 September 2026, the evaluators identified weak technical understanding and preparation, unclear features and project explanation, and presentation weaknesses including repeated pacing. They challenged the assumption that all stores want employees to approach customers frequently, citing luxury retailers such as Gucci and Prada. They considered the project worthwhile if its scope and explanation were properly focused. Appendix A reproduces the feedback text.

The response is to define a focused camera-based service-analytics scope, explain each component and its evidence, and distinguish proximity from an actual service interaction. Employee behaviour must be interpreted against the individual store's service policy. This report does not claim that one approach-frequency rule generalises to every retail format.

3.2 Proposal Action Register

Table 3.1: Proposal action register

No.	Feedback / Required Change	Response / Next Action	Status	Evidence
1	Improve technical understanding and preparation.	Document detection, classification and interaction rules; use executed tests to explain behaviour and limits.	Ongoing	Chapters 5, 6, 7

No.	Feedback / Required Change	Response / Next Action	Status	Evidence
2	Explain exact features and focus the project.	Focus scope on camera-based customer/staff analytics and the web application; make ownership and current feature status explicit.	Report updated	Chapters 1, 8
3	Defend applicability across different retail service models.	Require store-specific service thresholds and interpretation; evaluate low-contact as well as assistance-oriented scenarios.	Design response; evaluation planned	Chapter 4, FR-29
4	Improve presentation clarity and avoid pacing.	Rehearse a structured feature demonstration, assign speakers and practise stationary delivery.	Planned	Mid-evaluation preparation

3.3 Iteration Strategy

Development is iterative. The initial increment establishes the web application, secure access and camera-processing prototypes. Subsequent increments connect detection, classification, interaction and engagement outputs to the dashboard and evaluate them on store footage. Feature branches track implementation work. The evaluation milestones are listed below.

Table 3.2: Evaluation milestones

Milestone	Date
Proposal defence	11 September 2026
Mid evaluation	21 October 2026
Final evaluation window	23 November–4 December 2026

3.4 Iteration 1 Plan

Objectives: establish the application shell and data flows that later analytics plug into, and demonstrate person detection, role classification and interaction detection.

Selected features: manager and employee pages; role-aware authentication; reporting interface; alerts feed; system status page; person/role analysis and an interaction-detection pipeline (a floor-distance experiment was also built and later descoped).

Deliverables: running frontend, local application backend, person-analysis and interaction outputs, and demonstration evidence (Chapter 6).

3.5 Iteration 2 Plan

Objectives: replace prototype data with real agent output and evaluate it.

Selected features: Detection, Classification, Engagement (including interaction factors), and Trajectory agents; live data in the dashboard pages; the Insight Narration agent; and quantitative evaluation on labelled clips.

Deliverables: vision-component integration, evaluation report, and updated dashboard bound to real outputs (Chapter 9).

3.6 Work Breakdown and Responsibilities

Table 3.3: Work distribution

Work package	Description	Owner
Frontend and UX	Shared frontend development	Moiz and Ruhab
Customer/employee pose detection	Pose-detection work	Saqib
Person distance	Distance between people; head/hips/feet band handling when a full person position is rejected (the metric floor-distance experiment was descoped)	Moiz
Classification	Classification work	Ruhab
Testing	Detection and pipeline testing	Saqib
Report	Report preparation	Ruhab

The table records assigned areas, not a claim that every assigned feature has been integrated or validated. Completed-work evidence is distinguished in Chapter 8.

3.7 Progress Against Plan

Table 3.4: Planned versus actual progress at mid-report

Milestone	Planned	Actual
Web application (manager and employee)	Iteration 1	Implemented
Floor-distance CLI	Current prototype increment	Built, then descoped (`8c3a6fe`)
Interaction pipeline with pose judge and benchmark	Iteration 1	Implemented (local CLI)
Vision agents wired to dashboard	Iteration 2	In progress (prototype data)
Labelled accuracy evaluation	Iteration 2	Started: detector KPIs and one labelled clip; broader labelling pending

3.8 Delays, Deviations and Recovery Plan

Scope correction. Cross-camera association is reserved for the next phase and is not a mid-iteration deliverable. **Deviation.** The floor-distance prototype was descoped on 20 October because the moving-camera footage supported almost no floor projections; body-height-normalised distance with a pose judge replaced it. **Residual gap.** The dashboard's camera-derived metrics still use prototype data. **Recovery:** the first FYP-2 sprint binds the Detection and Engagement outputs to the dashboard, and quantitative evaluation is scheduled for the middle of the semester so that tuning time remains.

Chapter 4

Software Requirements

4.1 User Interactions / Use Cases

Table 4.1: Use case summary

ID	Use case	Actor	Module
UC-01	Sign up and log in (role-aware)	Manager, Employee	Web application
UC-02	View live camera dashboard with overlays	Manager	Dashboard
UC-03	View customer service-quality metrics	Manager	Customer Info
UC-04	View staff score, leaderboard, and idle time	Manager	Employee Info
UC-05	Explore heatmaps and movement patterns	Manager	Spatial Analytics
UC-08	Build and export a report (PDF)	Manager	Reports
UC-09	Review and filter alerts	Manager	Alerts
UC-10	View agent pipeline health	Manager	System Status
UC-11	View shift overview and personal performance	Employee	My Workspace
UC-13	Manage profile and notification preferences	Manager, Employee	Profile
UC-14	Calibrate and inspect floor-distance prototype (descoped)	Calibration operator / project team	Vision CLI

ID	Use case	Actor	Module
UC-15	Run interaction detection on a video	Project team / operator	Vision CLI

4.2 Use Case Diagram

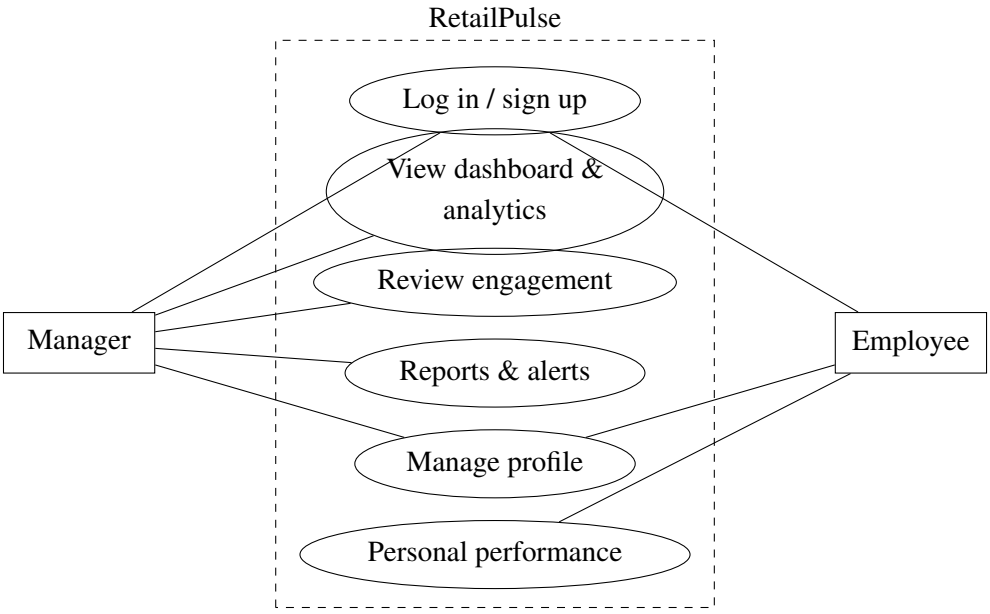


Figure 4.1: Use case diagram

4.3 Detailed Use Cases

UC-03: View customer service-quality metrics

Actor	Manager
Precondition	Manager is authenticated with a valid session.
Main flow	1. Manager opens Customer Info. 2. System shows footfall, attended versus unattended ratio, response time, and interaction duration. 3. Manager reviews dwell time, shopper-behaviour split, and top paths.
Alternative flow	No data for the selected interval: the system shows an empty state.
Postcondition	Metrics displayed; no data modified.

UC-14: Calibrate and inspect floor distance (descoped)

Actor	Calibration operator or project team.
Precondition	Local video and trained weights exist. Image/floor landmarks or explicitly estimated rectangle dimensions are available.
Main flow	1. Generate the uncalibrated foot-point preview. 2. Select or load ordered landmarks. 3. Save calibration and inspect the grid. 4. Run the distance demo and inspect video, floor map and records.
Alternative flow	Missing files, incompatible geometry, invalid calibration, unsupported foot positions or unverified alignment produce actionable errors or unavailable measurements.
Postcondition	Local diagnostic files and their configured measurement-status metadata are produced.

UC-14 belonged to the descoped floor prototype (Section 6.4.3). UC-14 and UC-15 are operator-facing CLI use cases and are separate from the manager/employee web interactions shown in the use case diagram.

UC-15: Run interaction detection on a video

Actor	Project team or operator.
Precondition	A local video, the trained role weights (`best.pt`) and the pose weights (`yolov8n-pose.pt`) exist.
Main flow	1. Run the interaction CLI on the video with the sampling options. 2. The pipeline detects and tracks employees and customers. 3. Proximity and duration nominate candidate pairs and the pass-by filter drops pairs that are walking past. 4. The pose judge confirms or rejects each candidate. 5. The CLI writes the annotated video and the JSON files.
Alternative flow	No pose found or the judge fails: the candidate is marked `unverified` and the proximity result stands. Judge switched off: all candidates are `unverified`.
Postcondition	Annotated MP4 with an INTERACTION ON/OFF indicator, and `.interactions.json`, `.rejected_interactions.json`, `.detections.json` and `.people.json` files.

Requirement and use-case identifiers are retained to preserve existing references; gaps in

numbering do not imply additional in-scope requirements.

4.4 Functional Requirements

4.4.1 Module 1: Vision Analytics

Table 4.2: Functional requirements: vision analytics

ID	Requirement	Priority
FR-01	The system shall detect persons in each processed frame.	High
FR-02	The system shall count footfall at the entrance camera per time interval.	High
FR-03	The system shall classify each tracked person as staff or customer without biometric data.	High
FR-04	The system shall compute approach time, interaction duration, and idle time.	High
FR-05	The system shall flag customers who remain unattended beyond a configurable threshold.	High
FR-06	The system shall compute an employee score from engagement metrics.	Medium
FR-07	The system shall record paths, zone dwell time, heatmaps, and crowd density.	High
FR-21	*(Descoped prototype.)* The system shall preview role labels, confidence, track IDs and box-bottom points without calibration.	High
FR-22	*(Descoped prototype.)* The system shall support ordered floor correspondences through saved configuration and a point-selection interface.	High
FR-23	*(Descoped prototype.)* The system shall calculate supported same-camera, same-timestamp pair separations in floor metres.	High
FR-24	*(Descoped prototype.)* Invalid or unsupported positions and distances shall have explicit states/reasons and unavailable values rather than fabricated zeroes.	High
FR-25	*(Descoped prototype.)* Camera alignment shall be monitored and shifted or unverified frames shall not produce supported metric output.	High

ID	Requirement	Priority
FR-26	*(Descoped prototype.)* Measurements, landmarks and operational coverage shall be editable so the homography can be regenerated.	High
FR-27	*(Descoped prototype.)* The system shall export annotated video, floor maps and timestamped diagnostic records.	Medium
FR-28	*(Descoped prototype.)* Estimated calibration and its distance outputs shall be labelled provisional and approximate.	High
FR-30	The system shall nominate an employee–customer pair as an interaction candidate when the pair stays within a body-height-normalised distance threshold (1.0, 0.8 or 0.6 body heights at the feet, hips or head band) for at least 2.0 seconds.	High
FR-31	A candidate whose relative position drifts by more than 0.5 body heights during the nomination window shall be treated as a pass-by and not reported as an interaction.	High
FR-32	A candidate shall be confirmed only if, in at least half of four pose crops from the last two seconds, a head points within 60° of the other person's head; otherwise it shall be rejected, or marked `unverified` when no pose is found or the judge fails.	High
FR-33	For each processed video the system shall write an annotated video with an INTERACTION ON/OFF indicator and JSON files for confirmed interactions, rejected interactions, detections and people.	Medium
FR-34	The system shall report detector, interaction and speed KPIs through a benchmark command.	Medium
FR-29	Engagement thresholds and unattended-customer interpretation shall be configurable for the store service policy; staff shall not be scored against a universal constant-approach expectation.	High

4.4.2 Module 2: Web Application

Table 4.3: Functional requirements: web application

ID	Requirement	Priority
FR-10	The system shall provide role-aware login and signup for Manager and Employee.	High
FR-11	The dashboard shall show camera feeds with overlays for footfall, engagement, and trajectory.	High
FR-12	Manager pages shall show customer, employee, and spatial analytics.	High
FR-13	The employee workspace shall show shift overview, daily score, 7-day trend, and zone assignment.	High
FR-17	The system shall build configurable reports with narrative summaries and export them as PDF.	Medium
FR-18	The system shall provide a filterable feed of active and resolved alerts.	Medium
FR-19	The system status page shall show per-agent health, throughput, and an activity log.	Low
FR-20	Users shall manage profile details, password, and notification preferences.	Low

4.5 Non-Functional Requirements

4.5.1 Reliability

NFR-01: Repeated geometric calculations using the same inputs and configuration shall agree within a defined numerical tolerance. NFR-02: Video-processing jobs shall report failure states and not fail silently. NFR-11 (descoped floor prototype): Floor projection shall reject incompatible frame geometry, stale homographies, invalid homogeneous denominators and unsupported operational coverage. Prediction and smoothing shall not hide alignment failure.

4.5.2 Usability

NFR-03: A manager shall reach any main analytics page within two navigation actions. NFR-04: Charts shall show textual values and use readable fonts (Space Grotesk, Inter, JetBrains Mono).

4.5.3 Performance

NFR-05: The vision pipeline shall run on a CPU-only machine, using frame sampling. NFR-06: Dashboard pages shall render within an acceptable interactive time on a typical laptop. No numeric response-time target has been agreed with the supervisor; report measured timings with their environment and scope.

4.5.4 Security

NFR-07: Manager-only application routes shall require a signed HttpOnly session cookie. NFR-08: Direct backend calls shall require an internal secret header, which the Next.js proxy supplies server-side. NFR-09: Secrets shall be configured through environment variables. NFR-10: No biometric identifiers shall be stored.

4.6 Acceptance Criteria

Table 4.4: Acceptance criteria for key requirements

Req.	Acceptance check
FR-03	Classification accuracy on labelled clips is measured and reported. No numeric pass threshold has been agreed; the supervisor accepted the 70-video dataset.
FR-05	A customer with no staff interaction beyond the threshold produces an alert in a controlled clip.
FR-01, FR-02	Detector precision, recall and mAP are reported on the held-out split (Table 7.4; TC-07).
FR-04, FR-30–FR-32	Interaction precision and recall are reported with and without the pose judge on the labelled clip (Table 7.5); a close pass-by is not reported as an interaction (TC-21, TC-22).
FR-33, FR-34	Every clip of the local set completes and writes the annotated video and JSON files; the benchmark metric functions return correct precision, recall and F1 (TC-23, TC-24).
FR-07	Zone dwell on a labelled clip matches the annotation within tolerance (TC-10, pending).
FR-10	A user with the Employee role cannot open manager pages.
FR-11–FR-13	Pages render for both roles and the Employee role is blocked from manager pages (TC-12).
FR-17	Exported PDF opens and contains the selected sections and narrative.
NFR-01	Two runs with identical inputs give byte-identical JSON outputs (TC-25).
NFR-05	Processing speed is reported as a realtime factor per clip on a CPU-only machine (Table 7.6).
NFR-07, NFR-08	Direct calls without the secret or cookie are rejected.

4.6.1 Floor-Distance Acceptance (Descoped Prototype)

1. **Software correctness (FR-21–FR-28, NFR-11):** known synthetic projections and distances agree with expected geometry; invalid cases return an error or unavailable state; illustrative configuration cannot silently serve as actual calibration.

2. **Integration:** local trained-model/video execution produces readable annotated

video, floor maps and timestamped records. Unsupported positions and shifted or unverified frames do not produce supported distances.

4.7 Requirements Traceability

Table 4.5: Traceability matrix

Req.	Module	Design / implementation	Test case
FR-01–FR-02	Detection Agent	YOLO detector, entrance camera	TC-07
FR-03	Classification Agent	Trained role detector in the interaction CLI; earlier heuristic demo	TC-08
FR-04–FR-06	Engagement Agent	Engagement metrics (pending integration)	TC-09
FR-30–FR-32	Interaction pipeline	Body-height proximity, pass-by filter, pose judge	TC-21, TC-22, TC-24
FR-33	Interaction CLI	Annotated video and JSON outputs	TC-24, TC-25
FR-34	Benchmark	Detector, interaction and speed KPIs	TC-23
FR-07	Trajectory Agent	ByteTrack paths, dwell, heatmap	TC-10
FR-29	Engagement policy	Store-specific configuration and policy-aware interpretation (planned)	Policy scenario evaluation
FR-10	Web application	Login, signed session	TC-01, TC-02
FR-11–FR-13	Web application	Next.js pages	TC-12
FR-17	Reports	jsPDF export	TC-13
FR-18–FR-20	Web application	Alerts, status, profile	TC-12

Table 4.6: Floor-distance requirements traceability (descoped prototype)

Requirement	Component	Implementation / design	Test case
FR-21	*(Descoped prototype.)* Detector / preview	Shared trained inference, ByteTrack, original-pixel foot proxy	TC-14
FR-22, FR-26	Calibration	Landmark JSON, point UI and regenerable homography	TC-15, TC-16
FR-23, FR-24, NFR-11	Floor geometry	Supported projection, unit/region guards and explicit failure states	TC-17, TC-18
FR-25	*(Descoped prototype.)* Motion monitor	Masked ORB matching, RANSAC and alignment-quality gates	TC-19
FR-27, FR-28	Distance demo	Video/map/JSONL exports and provisional metadata	TC-20

Chapter 5

System Design

5.1 System Architecture

The mid-iteration design separates person detection, role classification, tracking, interaction judging and trajectory analysis. The Next.js frontend presents manager and employee views. The diagram shows component responsibilities; it does not imply that all vision outputs are already connected to the dashboard.

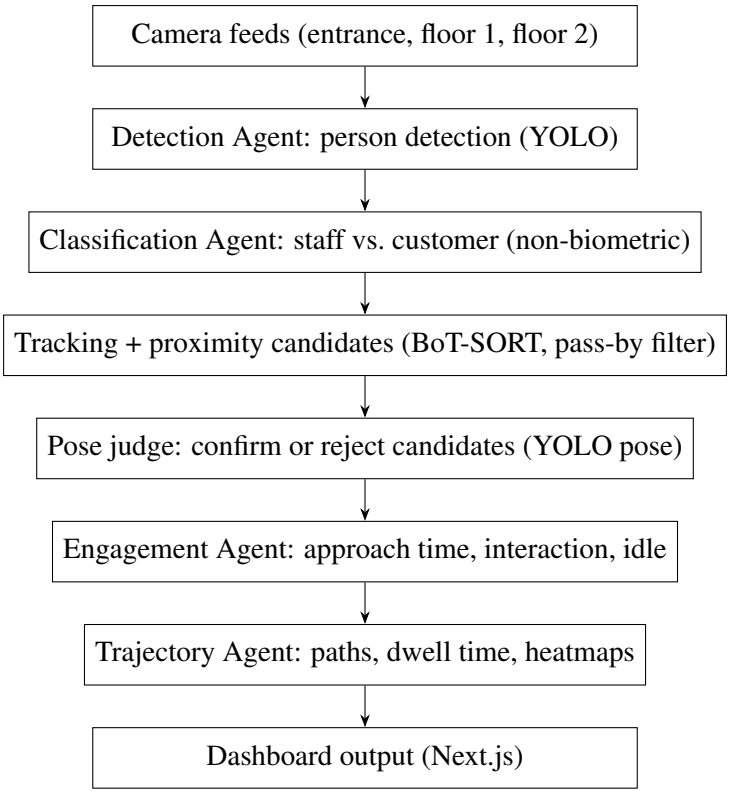


Figure 5.1: RetailPulse target multi-agent pipeline; tracking, proximity candidates and the pose judge run in the interaction CLI and their downstream integration is pending

The application tier is separate from the vision pipeline. The Next.js frontend serves the pages and proxies backend calls. A local FastAPI service supports application records and exposes `/camera/*` for camera-processing jobs.

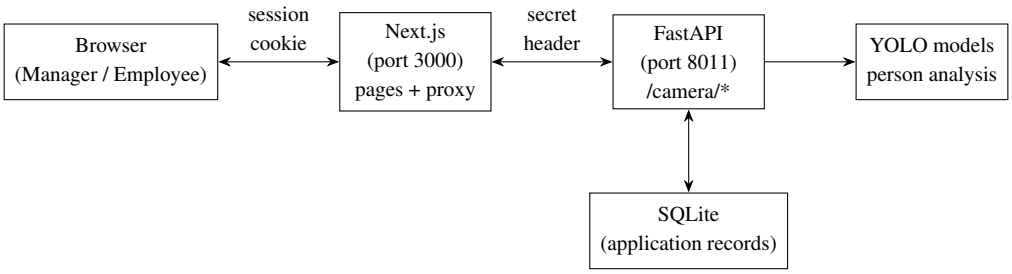


Figure 5.2: Application tier and service boundaries

5.1.1 YOLO Model Responsibilities

The project uses two distinct YOLO model roles: a YOLO pose model supplies body-keypoint evidence for interaction analysis, while a standard YOLO object-detection model with customer/employee classes supplies person boxes and role labels. The latter

is a detector trained on role classes, not a separate image-classification-only network. Interaction events are derived from pose/proximity and temporal rules rather than treating a pose prediction itself as an interaction label.

The repository contains two related execution paths. In the interaction CLI, `InteractionPipeline` uses the role detector and bounding-box body bands to nominate candidate pairs, and the pose model then judges each nominated pair (Section 6.4.2). Separately, `CVModels` in `backend/interaction/models.py` supports a detector and an optional pose checkpoint, and `FrameProcessor` matches pose boxes to tracked boxes and derives orientation evidence from keypoints. The head/hips/feet fallback is a geometric band-selection rule; it is not a third YOLO model and does not establish anatomical keypoint visibility. When the judge is switched off the pipeline runs on the role detector alone and candidates are reported as `unverified`.

5.2 Architectural Components

Table 5.1: Architectural components

Component	Description
Frontend	Next.js (App Router) and React pages, Tailwind CSS styling, Recharts charts, and jsPDF export. UI primitives live in `src/components/ui`.
Proxy layer	Next.js server-side routes that forward calls to FastAPI and attach the internal secret.
Vision agents	Detection, Classification, Engagement, and Trajectory agents; downstream web integration remains in progress.
Interaction pipeline	Role-labelled detections, BoT-SORT tracking, body-height proximity with a pass-by filter, and a YOLO pose judge; runs as a local CLI and writes video and JSON outputs.
Narration agent	Produces natural-language summaries for reports.

5.3 Interfaces and Dependencies

- Browser to Next.js: HTTP, with a signed HttpOnly session cookie for manager routes.

- Next.js to FastAPI: HTTP on `127.0.0.1:8011`, carrying an internal secret header added server-side.
- FastAPI to SQLite: local database files for accounts, cameras, alerts, tasks and interaction documents.
- Interaction CLI to filesystem: a local video and the model weights produce an annotated MP4 and JSON files (`.interactions.json`, `.rejected_interactions.json`, `.detections.json`, `.people.json`). Dashboard integration of these outputs is pending.
- Configuration: session-signing and internal-service secrets are supplied through environment variables and held server-side.
- Key dependencies: Node.js 18+, Python 3.12+, Ultralytics YOLO, OpenVINO, BoT-SORT/ByteTrack, Recharts, jsPDF.

5.4 Architecture Decisions and Rationale

Table 5.2: Architecture decisions

Decision	Rationale
CPU-only, lightweight YOLO	Matches hardware constraints and the target retailer.
Non-biometric classification	Privacy-conscious design and simpler compliance.
Next.js proxy with server-held secret	Keeps secrets and role checks out of the browser.
SQLite for application records	Zero-administration storage that suits a single-store prototype.
Offline/batch processing	Real-time guarantees are not feasible on CPU for the committed scope.
Body-height-normalised proximity	Needs no calibration and works on moving-camera footage; it replaced a calibrated floor homography that supported almost no projections.
Pose judge on proximity candidates	Separates a close pass-by from a face-to-face interaction; the verdict stays `unverified` when no pose is available.

5.5 Design Evolution

The mid-iteration scope is focused on person/role analysis, interaction detection and the web application, using offline/batch video processing. Three changes since the proposal matter.

1. Authentication was upgraded from client-side role flags to server-signed session cookies and an internal secret header.
2. A calibrated floor-homography prototype (commit `4a9023a`) was built to measure person separation in metres. It was descoped and removed (commit `8c3a6fe`) because the moving-camera footage supported almost no floor projections (Section 6.4.3).
3. Body-height-normalised proximity replaced it, and a YOLO pose judge was added (ported in commit `6560403`) because proximity alone reported close pass-bys as interactions (Table 7.5).

None of this creates a common cross-camera coordinate system; cross-camera association remains next-phase work.

5.6 Domain Model

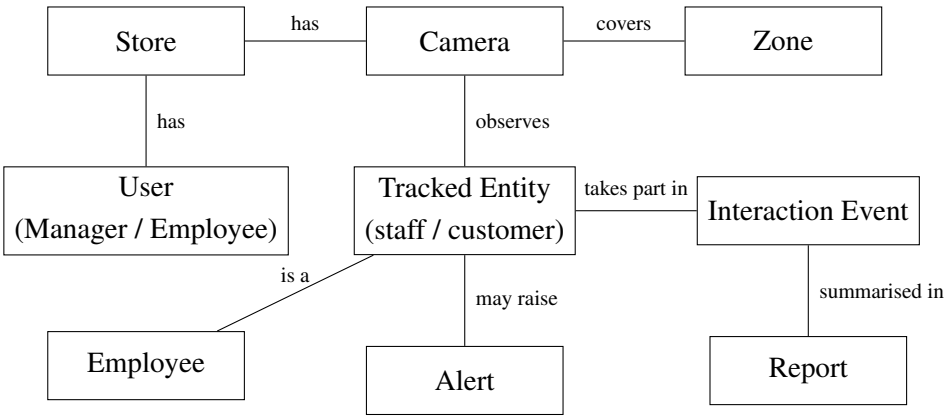


Figure 5.3: Conceptual domain model

5.7 Data Design

The following selected in-scope schema was verified against the branch on 20 October 2026 at commit `52f651e`. Commit `6560403` (pose judge and benchmark) was added

afterwards and this table has not been re-checked against it.

Table 5.3: Implemented persistence and exported record structure

Source / table	Actual fields or structure
backend/store_core.py: users	id, name, email, role, password_hash, active, created_at, updated_at
invitations	id, email, role, token, expires_at, accepted_at, created_by, created_at
zones	id, name, type, capacity, min_staff, created_at, updated_at
cameras	id, name, zone_id, role, source_type, source_url, online, created_at, updated_at
rosters	id, user_id, zone_id, shift, available, created_at, updated_at
settings; meta	key, value_json, updated_at; key, value respectively.
alerts	id, category, severity, title, context, href, status, source, evidence_json, created_at, resolved_at, resolved_by
dispatch_decisions	id, event_id, alert_id, task_id, selected_employee_id, decision, reason, policy_json, created_at
tasks	id, employee_id, target_zone_id, source_zone_id, instruction, state, trigger_alert_id, evidence_json, created_at, updated_at, due_at
task_events	id, task_id, state, actor_id, note, created_at
backend/interaction/storage.py: documents	kind, store_id, id, data; composite primary key on the first three columns; data contains JSON.
jobs	id, store_id, status, created_at, data; index job_store on store_id, created_at.
Tracked person	track_id, bbox, confidence, role, role_confidence, role_evidence; serialised centre derived from the box.

Source / table	Actual fields or structure
Interaction event	`employee_track_id`, `customer_track_id`, `started_at`, `last_seen_at`, `distance_body_heights`, `measured_by`, `active`, `duration_seconds`.

The standalone interaction distance is dimensionless body-height-normalised image separation. It must not be interpreted as the metres in the historical floor-homography records below. Figure E.1 shows selected implemented relationships.

5.7.1 Floor Calibration and Runtime Records (Descoped Floor Prototype)

Versioned calibration JSON stores camera/calibration IDs, the video reference and timestamp, original frame dimensions, the coordinate-space definition, metre units, image/floor landmarks, an image-to-floor homography, operational/exclusion polygons, distortion metadata, motion/smoothing settings and fitting diagnostics. Measurement-status metadata identifies estimated calibration inputs.

The local editable input is `data/calibration/queue_temp.landmarks.json`; generated calibration is `data/calibration/queue_temp.json`. The authoritative rectangle fields and original-frame landmark pixels are used to regenerate the matrix. A modified input with a stale stored matrix is rejected. Non-sensitive examples and the schema are stored under `config/cameras/`; footage, real calibration and diagnostics are excluded from Git.

Runtime JSONL records store frame index, actual decoded timestamp, camera/calibration identity, track ID, role, joint detection/class confidence, pixel box and foot point, supported floor position, motion diagnostics, pair distances and explicit rejection reasons. Unavailable positions/distances are JSON `null`. `observed_proxy` denotes an accepted box-bottom estimate, not verified foot visibility; `predicted` denotes short extrapolation and is excluded from supported pair measurements.

5.8 Process / Behavior Design

In the active standalone `backend/interaction/pipeline.py` path, `InteractionConfig` and `engagement.py` define engagement between employee/customer track pairs. The lowest common visible box band is selected: feet (65–100% of box height), hips (25–70%), then head (0–30%). The edge-to-edge Euclidean band gap is divided by mean

pair box height. The default thresholds are 1.0, 0.8 and 0.6 body heights respectively, with a minimum apparent-height ratio of 0.6. Frame-edge truncation and nearer-person occlusion can invalidate bands; the edge margin is four pixels.

The candidate pair must persist for at least 2.0 seconds; a missing gap above 1.5 seconds ends continuity. This standalone proximity stage imposes no facing-angle test; in the interaction CLI it is followed by a pass-by filter and a pose judge (Section 6.4.2). Its defaults use the trained OpenVINO role model, three-second role retention and the configured BoT-SORT/ReID tracker. A separate frame-processing path has other interaction settings; those settings are not substituted for these standalone defaults. Proximity is an operational engagement heuristic, not proof of conversation or service. The proposed unattended-customer alert flow requires separate integration evidence.

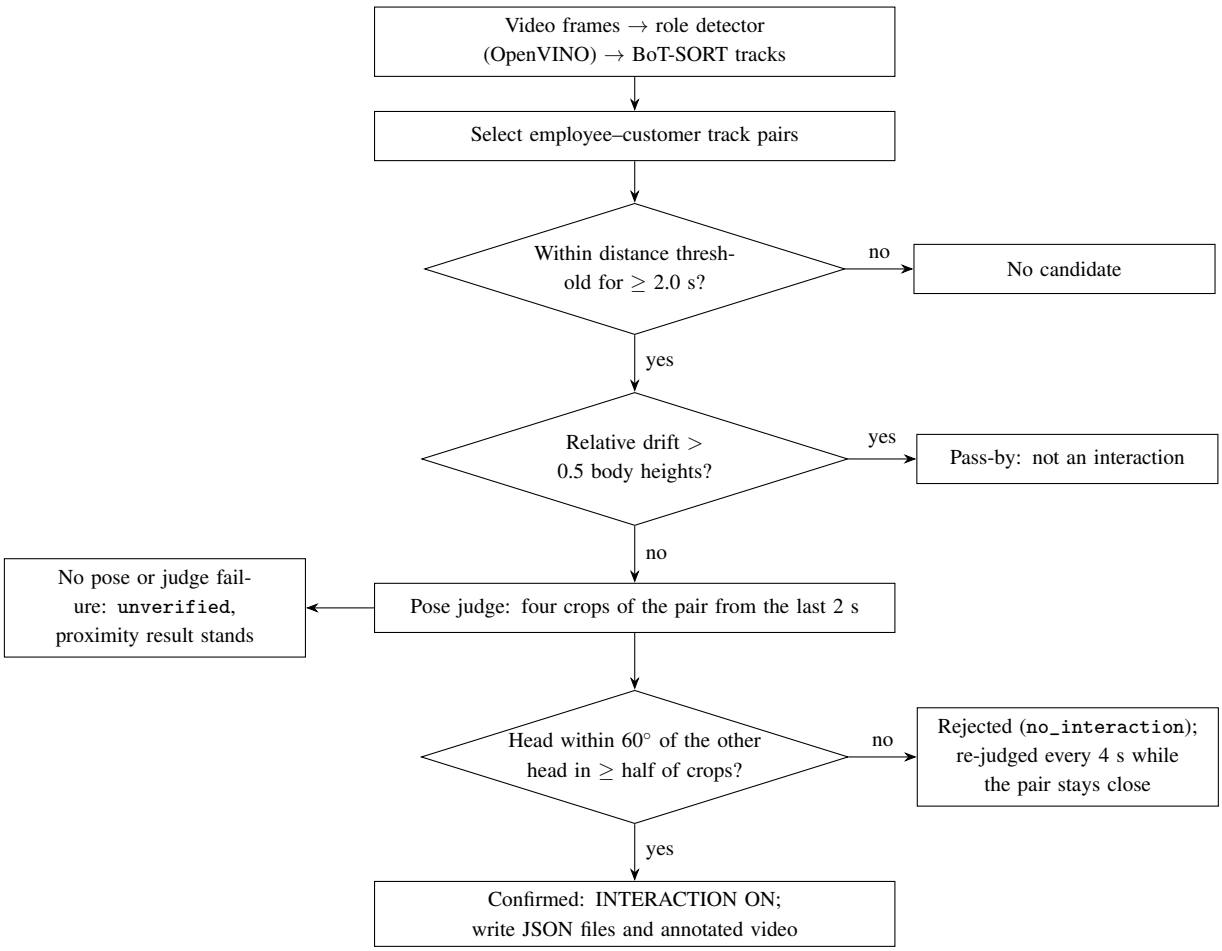


Figure 5.4: Interaction detection process in the interaction CLI

Figure 5.4 summarises the interaction decision flow. Candidate nomination uses only boxes and tracks; the pose judge is the only stage that uses body keypoints.

5.8.1 Floor-Distance Quality-Gating Process (Descoped Prototype)

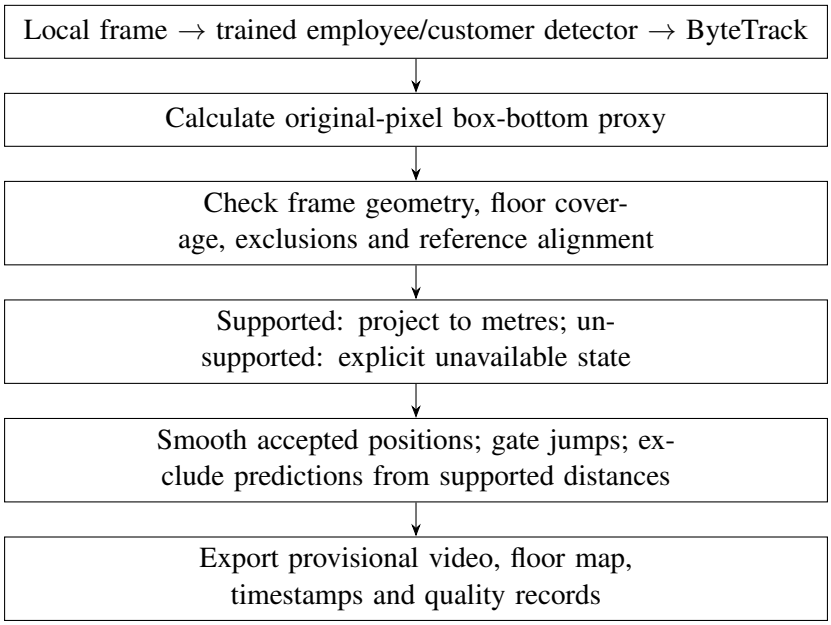


Figure 5.5: Floor-distance localization and quality-checking pipeline

The reference frame supports a provisional snapshot. Other frames require sufficient distributed static floor evidence within configured tolerances. A lack of matches is unverified alignment, not a stationary-camera decision. The current implementation monitors and rejects movement; it does not compensate for it. Metric smoothing cannot repair camera mapping errors. Trained roles, body-height proximity, duration and the pose judge are now combined in the interaction CLI; dashboard integration remains a future step, and proximity alone is not proof of assistance.

Chapter 6

Iteration 1: Design and Implementation

6.1 Iteration Objectives

Build the application platform and data services that later vision analytics plug into, secure it, and demonstrate person/role analysis and interaction detection.

6.2 Selected Use Cases / Features

UC-01–UC-05, UC-08–UC-11, UC-13 and the operator-facing UC-14 (Chapter 4), covering the application and person-analysis features and the additional FR-21–FR-28 floor-distance prototype. The floor-distance prototype has since been descoped and removed from the current branch (commit `8c3a6fe`); its sections below are kept as historical evidence.

6.3 Design

The design follows Chapter 5: a Next.js frontend with a server-side proxy, a FastAPI service for application and `/camera/*` workflows, SQLite for application records, and local person-distance diagnostics.

6.4 Implementation

The frontend follows the App Router layout, with one folder per page (landing, login, signup, dashboard, customer-info, employee-info, spatial-analytics, alerts, settings, employee-home, my-performance, profile, about, camera-handoff, compare-people,

voice-agent). Shared UI primitives (RetroWindow, RetroCard, RetroButton, Navbar, Footer) live in `src/components/ui`, and session-signing helpers live in `src/lib`. The backend lives in `backend/`.

6.4.1 Two-Model Vision Approach

The project uses two distinct YOLO model roles: a YOLO pose model supplies body-keypoint evidence for interaction analysis, while a standard YOLO object-detection model with customer/employee classes supplies person boxes and role labels. The latter is a detector trained on role classes, not a separate image-classification-only network. Interaction events are derived from pose/proximity and temporal rules rather than treating a pose prediction itself as an interaction label.

In the interaction pipeline the trained customer/employee detector (`best.pt`, run as an OpenVINO export on CPU) supplies role-labelled boxes, and the pose model (`yolov8n-pose.pt`) acts as an interaction judge: once distance and duration rules nominate an employee/customer pair, it confirms the pair only if someone's head is turned towards the other person. These responsibilities are distinct from tracking (BoT-SORT with appearance re-identification) and from the distance/duration rules that nominate candidates. The head/hips/feet approximation used when a full-person position is rejected belongs to the distance logic and should not be confused with employee/customer role classification.

6.4.2 Interaction Pipeline and Pose Judge

`python-mbackend.interaction.cli` runs the interaction pipeline on a video. Interaction detection has two stages. Proximity nominates a candidate: an employee and a customer must stay within a body-height-normalised distance (1.0, 0.8 or 0.6 body heights measured at the feet, hips or head band) for at least two seconds, and a customer whose position relative to the employee drifts by more than 0.5 body heights in that time is treated as walking past. The pose judge then takes four crops of the pair from the last two seconds and confirms the candidate only if, in at least half of them, someone's head (nose relative to the ears) points within 60° of the other person's head. If no pose is found or the judge fails, the verdict is `unverified` and the proximity result stands; a rejected pair is re-judged every four seconds while it stays close. For each video the CLI writes an annotated MP4 with an INTERACTION ON/OFF indicator, `.interactions.json` (confirmed), `.rejected_interactions.json` (vetoed), `.detections.json` and `.people.json`. `python-mbackend.interaction.benchmark` measures detector, interaction and speed KPIs (Chapter 7).

Running the interaction pipeline

Run every command from the repository root with the Python virtual environment active (Python 3.12+). The role weights `models/best.pt` and the pose weights `yolov8n-pose.pt` must be present locally; they are not stored in Git. Footage is also kept out of Git.

```
python -m backend.interaction.cli <path-to-video>
  -every 10 -track-every 3
python -m backend.interaction.benchmark
  backend/interaction/benchmarks/benchmark.json
python -m pytest backend/interaction -q
PYTHONPATH=backend python -m pytest backend/test_handoff_core.py
-q
```

`-every 10` scores every tenth frame and `-track-every 3` tracks every third frame, as used for the 11-clip run in Chapter 7. The tests report 44 passed for the interaction suites and 5 passed for the camera-handoff core.

6.4.3 Floor-Distance Command-Line Prototype (Descoped)

This prototype was implemented in commit `4a9023a` and removed in commit `8c3a6fe`; the files named in this and the following floor sections exist only in that history. It was descoped because the available moving-camera footage supported almost no floor projections, and body-height-normalised distance replaced it. The implementation reused the existing local trained `models/best.pt` weights, whose classes are employee and customer, without retraining or replacement weights. Shared loading/tracking helpers in `backend/vision_inference.py` provide the YOLO/ByteTrack inference route for the floor-distance prototype. The floor CLI selects both role classes; it does not substitute the earlier vertical-position role heuristic.

The entry points are `scripts/preview_floor_points.py`, `scripts/calibrate_floor.py`, `scripts/run_floor_distance.py`, and `scripts/evaluate_floor_distance.py`. Geometry, calibration diagnostics, motion monitoring, metric smoothing and evaluation are separate backend modules. The preview runs without calibration; the distance demo exports only supported metre observations and explicit unavailable states. No frontend changes were needed for this prototype.

6.5 Completed Features / Module

Table 6.1: Features completed in Iteration 1

Feature	Detail	Status
Manager dashboard pages	Dashboard, Customer Info, Employee Info, Spatial Analytics, Alerts, System Status, Settings	Done (prototype data for vision metrics)
Employee workspace	My Workspace and My Performance with rubric and daily log	Done (prototype data)
Shared pages	Landing, Login/Signup, About, Profile	Done
Reports	Configurable builder, narrative summary, PDF export	Not on tested branch (route returns 404)
Secure access	Stored accounts (PBKDF2 password hashes), signed HttpOnly session cookie and internal secret header	Done
Floor-distance prototype	Trained roles/track IDs, foot preview, calibration, alignment checks and diagnostic exports	Implemented, then descoped (`8c3a6fe`)
Interaction pipeline	Trained role detector, BoT-SORT tracking, body-height proximity, pass-by filter and YOLO pose judge; annotated video and JSON outputs	Implemented (local CLI)
Benchmark	Detector, interaction (with and without the judge) and speed KPIs	Implemented

6.6 Technical Implementation Details

Running the system. `npm run dev:all` starts Next.js on port 3000 and FastAPI on `127.0.0.1:8011`. `npm run backend:api` starts the application backend and its `/camera/*` routes. The package scripts use Unix-style environment assignment; Windows users should use the documented platform-compatible launcher or set environment

variables explicitly.

Authentication. Login verifies stored account credentials and creates a server-signed HttpOnly session cookie. Protected manager backend routes require the signed session and server-side role checks. Internal secrets are supplied through environment variables. The executed access-control checks are recorded in Chapter 7.

6.6.1 Calibration Configuration and Coordinate Conventions (Floor Prototype)

Detections use original-frame $[x_1, y_1, x_2, y_2]$ pixels. Ultralytics already restores xyxy boxes after preprocessing, so coordinates are not unletterboxed again. Frame/configuration dimension mismatches, invalid projections and stale homographies are rejected. No lens undistortion is applied; calibration and runtime share the same distorted pixel space. Intrinsic/distortion metadata slots are retained, but correction would require an explicitly consistent update of both paths.

The provisional rectangle follows A–B–D–C: AB = DC = 3.13225 ft and BD = CA = 4.2075 ft, equivalent to width 0.9547098 m and depth 1.282446 m. The configuration records the rectangle geometry and estimated lengths. Dot centres in the 479-by-479 marked screenshot were mapped to the original 1080-by-1080 video using pixel-centre resize coordinates. Frame zero was selected after image-difference comparison and visual inspection; the resized/modified screenshot correspondence remains approximate. Operational coverage is restricted to the marked rectangle.

The rectangle dimensions and original-pixel points are editable in configuration. Regeneration writes the image-to-floor matrix and fitting diagnostics without changing trained weights. Saved measurements are labelled estimated, calibration is provisional and runtime values are approximate. Point-click selection supports numbered correspondences, undo, reset and save; the headless JSON route was exercised.

6.6.2 Reproducible Execution (Floor Prototype)

These commands ran at commit `4a9023a`; the scripts are not on the current branch. From the repository root, use the existing virtual environment. The following three command groups reproduce the sampled temporary-video workflow; join wrapped continuation lines when entering each command:

```
.venv/Scripts/python.exe scripts/preview_floor_points.py
  -video data/videos/raw/queue_temp.mp4 -sample-every 30 -headless
.venv/Scripts/python.exe scripts/calibrate_floor.py
```

```
  -video data/videos/raw/queue_temp.mp4 -timestamp 0
  -landmarks data/calibration/queue_temp.landmarks.json
  -output data/calibration/queue_temp.json -headless
.venv/Scripts/python.exe scripts/run_floor_distance.py
  -video data/videos/raw/queue_temp.mp4
  -calibration data/calibration/queue_temp.json
  -motion-mode monitor -sample-every 30 -headless
```

The displayed line breaks are for typesetting; join continuation lines when entering a command. The output directory contains annotated video, top-down video, JSONL observations and summary/coverage records. Actual decoded timestamps are exported. Repository usage instructions are in `docs/floor_distance_usage.md`.

6.7 Algorithms and Technical Decisions

6.7.1 Person Floor-Position Approximation (Floor Prototype)

For a box $B = (x_1, y_1, x_2, y_2)$, the homogeneous floor-contact proxy is

$$\mathbf{p} = \begin{bmatrix} (x_1 + x_2)/2 & y_2 & 1 \end{bmatrix}^{\mathsf{T}}. \tag{6.1}$$

This is an observed proxy rather than proof that feet are visible. Boundary truncation, person overlap and manual exclusion polygons provide provisional rejection signals; seated people and chair occlusion remain limitations. An optional future foot/keypoint locator can replace this step without changing the projection interface.

6.7.2 Homography-Based Projection and Distance (Floor Prototype)

An image-to-floor homography maps the proxy into metre coordinates [11, 12]:

$$\mathbf{q} = H_{\text{image}\to\text{floor}}\mathbf{p}, \qquad X = q_1/q_3, \quad Y = q_2/q_3. \tag{6.2}$$

Non-finite values or a denominator near zero are rejected. For two supported observations from the same camera and timestamp,

$$d_{AB} = \sqrt{(X_A - X_B)^2 + (Y_A - Y_B)^2} \quad \text{metres}. \tag{6.3}$$

Four non-degenerate landmark pairs provide a direct calibration fit. With additional correspondences, the implementation fits floor-to-image using RANSAC with an explicit pixel-

space threshold, then inverts the result [14]. Inliers and fitting residuals are exported as calibration diagnostics.

6.7.3 Camera Alignment and Measurement Reliability (Floor Prototype)

The monitor compares ORB features [15] within the supported floor region, masking detected people and manual exclusions. Matches are checked using RANSAC, inlier count/ratio, spatial spread, residual displacement and supported-region displacement limits. Insufficient evidence produces `motion_unverified`; supported alignment shifts produce `motion_invalid`. Neither state allows a supported metre output. Modes `monitor` and `fixed` both enforce these checks; `snapshot` supports only the reference frame. Motion compensation is deferred because reliable distributed floor evidence was not established on the temporary clip.

Accepted positions are smoothed in metres using timestamps. Configurable speed/jump gates, short prediction limits, a separate last-reliable-observation time and track expiry preserve quality information. Predictions are excluded from supported pair distances, and alignment failures clear smoothing state. Role labels remain separate observations from the trained detector; their confidence is the joint detection/class score, not two independent probabilities. Selected track pairs or a capped set of all pairs can be exported.

6.8 Tools, Frameworks and Technologies

Table 6.2: Technology stack

Layer	Technology
Frontend	Next.js (App Router), React
Styling	Tailwind CSS, custom retro-industrial design system
Charts	Recharts
Computer vision	YOLO pose model as interaction judge; trained YOLO role detector (OpenVINO, CPU), BoT-SORT with ReID, OpenCV, NumPy; `lap` for tracking assignment; OpenCV homography/ORB in the descoped floor prototype
Application and camera API	FastAPI, SQLite
Fonts	Space Grotesk, Inter, JetBrains Mono
Runtime	Node.js 18+, Python 3.12+ (virtual environment)

6.9 Demonstration Evidence

Local UI screenshots were captured on 20 October 2026 using isolated test databases. The displayed analytics are demonstration data, not measured vision accuracy. The PDF-report route returned HTTP 404 on the tested branch, so no screenshot is claimed for that feature.

6.9.1 Interaction Pipeline Output

Figure 6.5 shows annotated output of the current interaction CLI on two local clips.

6.9.2 Floor-Distance Demonstration (Descoped Prototype)

Figures 6.6–6.8 show actual local prototype outputs from commit `4a9023a`. They demonstrate inference, calibration visualization and export.

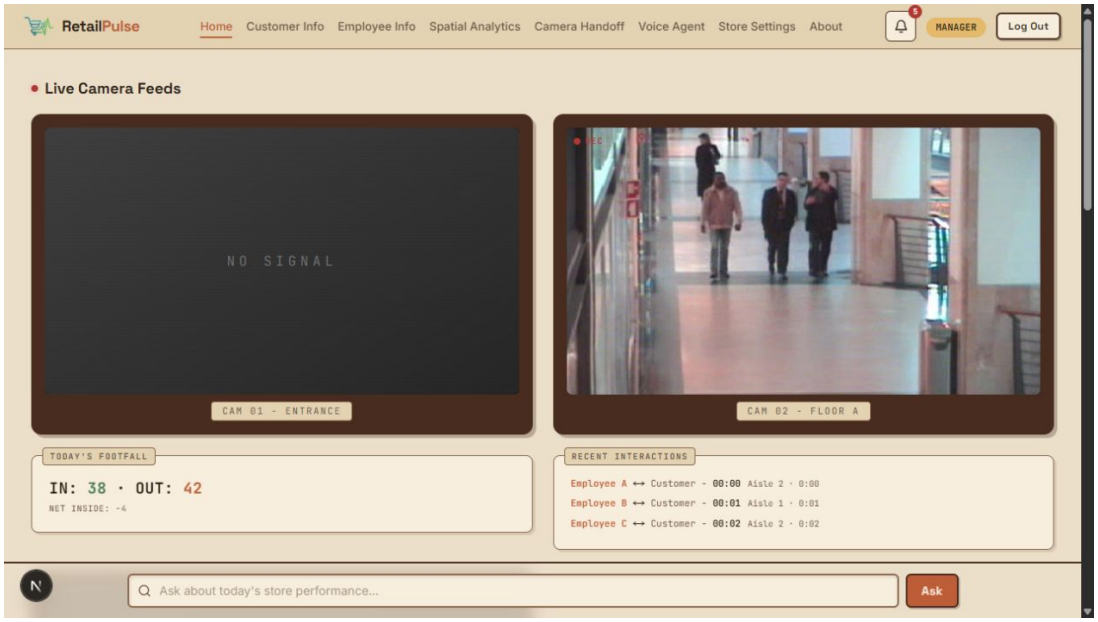


Figure 6.1: Manager dashboard with demonstration values

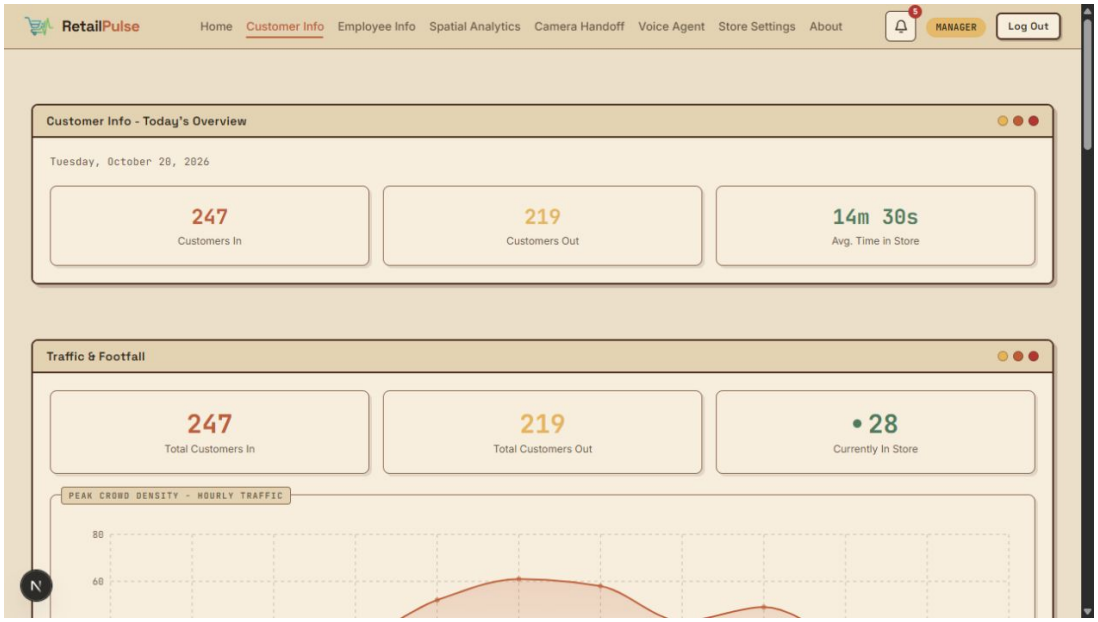


Figure 6.2: Customer information view with demonstration values

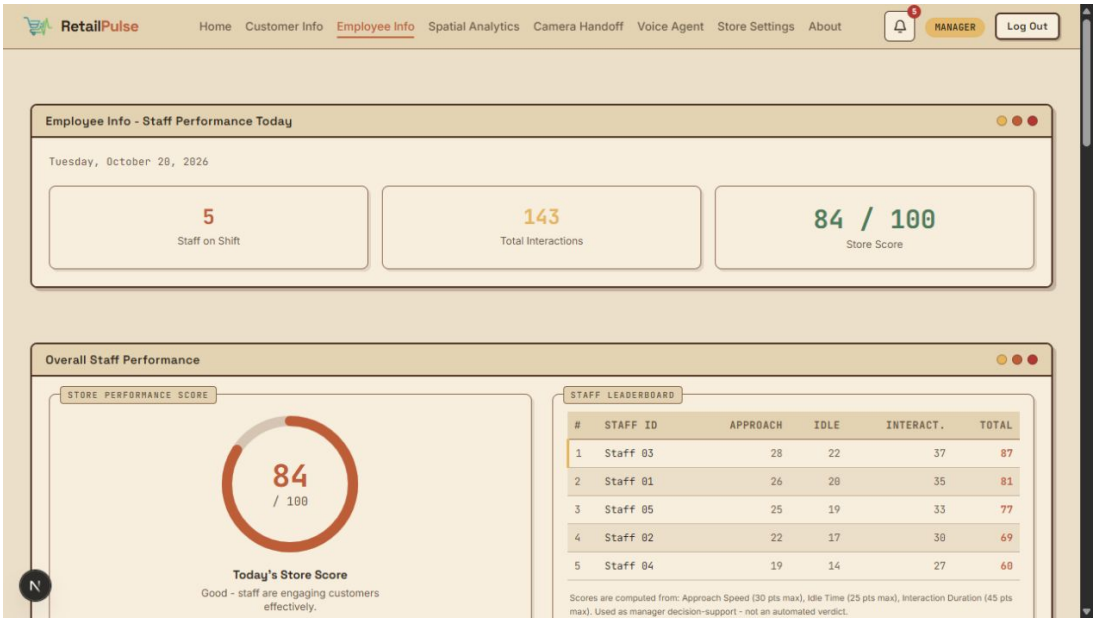


Figure 6.3: Employee information view with demonstration values

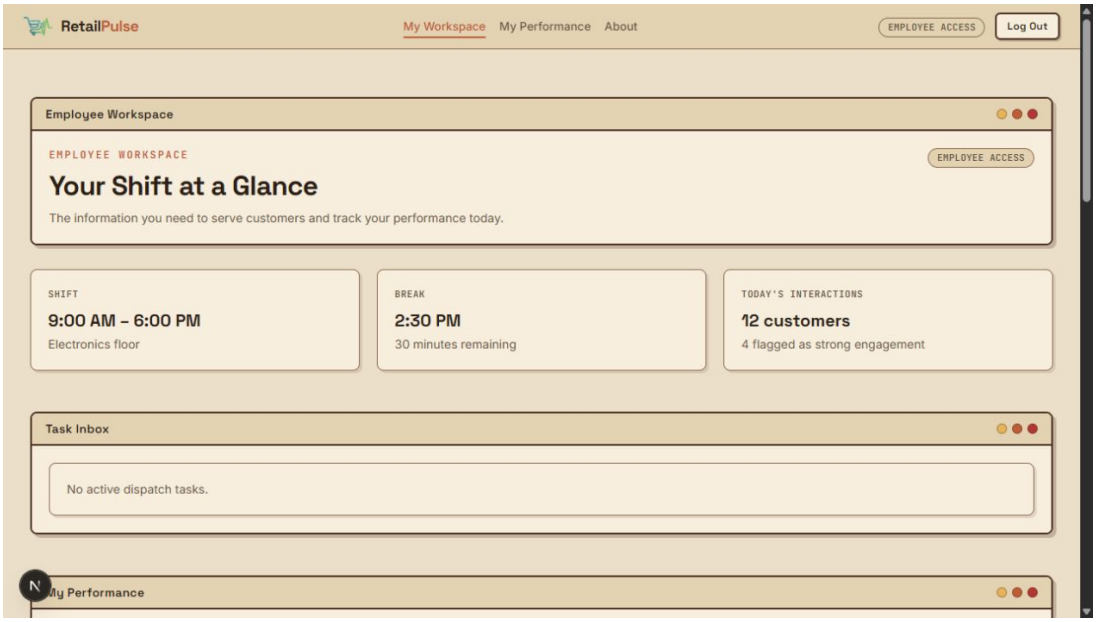


Figure 6.4: Employee workspace after authenticated employee login

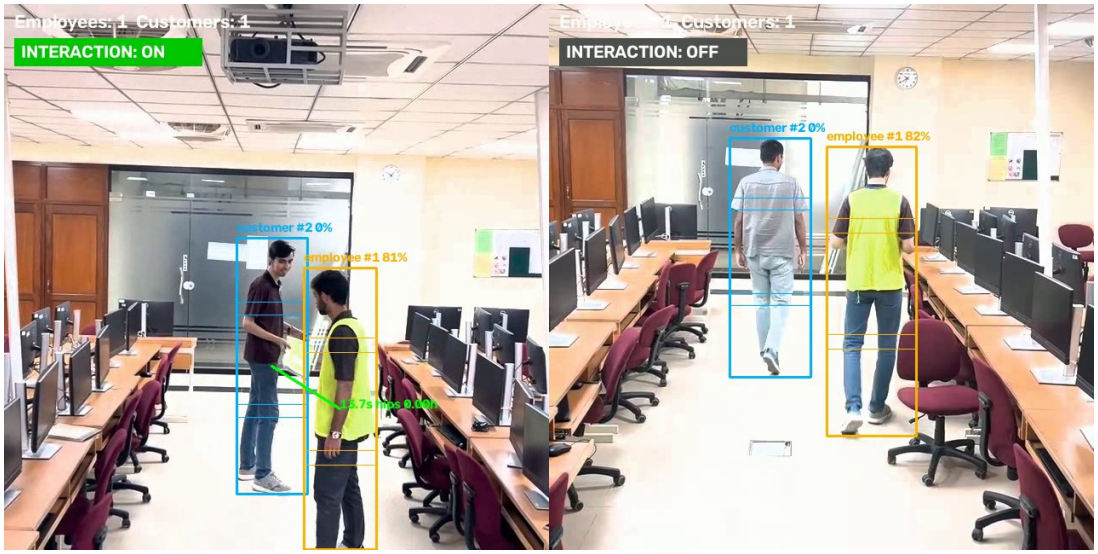


Figure 6.5: Interaction pipeline output. Left (IMG_8880, 30 s): employee and customer facing each other during a handover; the pose judge confirms the interaction. Right (IMG_8829, 8 s): employee and customer close together but walking away side by side; the pose judge rejects the candidate and the indicator stays OFF.

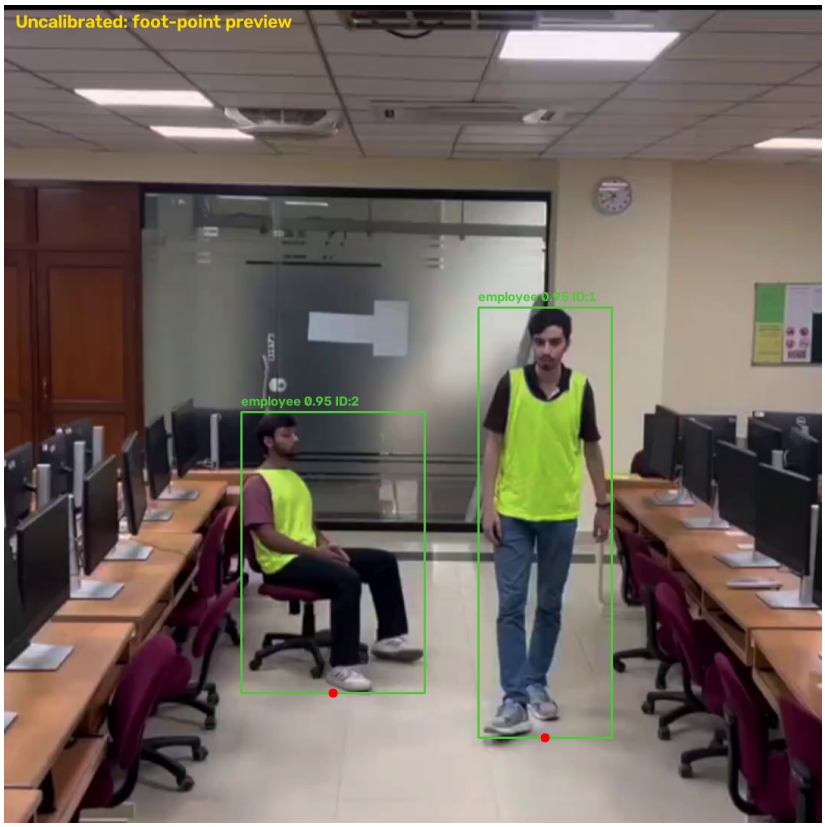


Figure 6.6: Trained role detections and track IDs with box-bottom proxies; uncalibrated preview

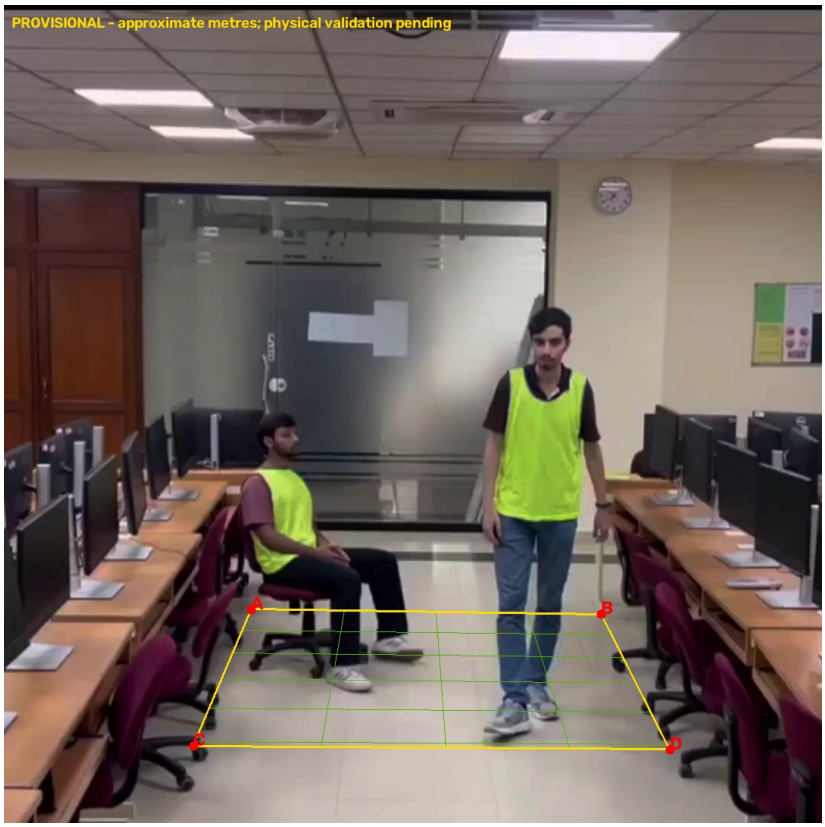


Figure 6.7: Provisional A–B–D–C calibration and floor grid based on estimated dimensions

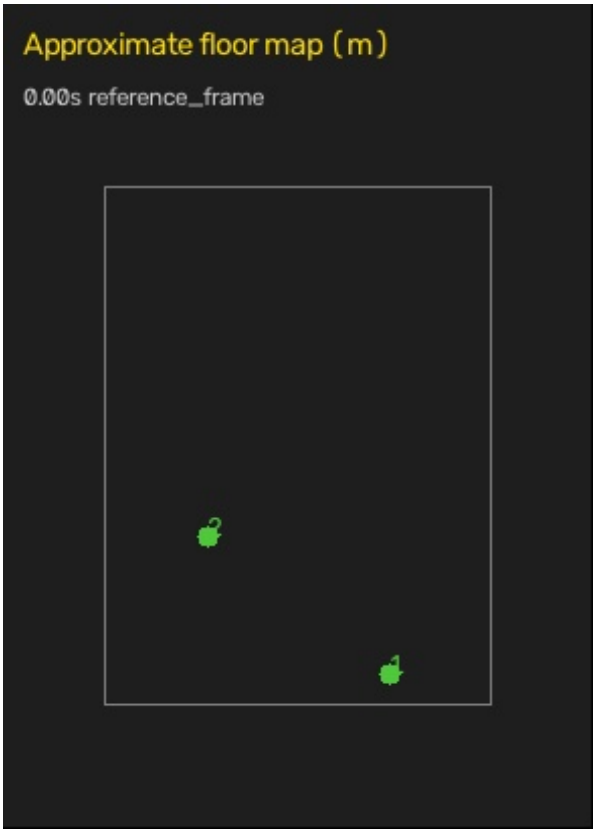


Figure 6.8: Approximate top-down person-position proxies in the reference snapshot

6.10 Iteration Outcome

Iteration 1 covers the application platform, role-aware access, person/role analysis, interaction cues and the distance-estimation prototypes. The vision-derived dashboard metrics still use prototype data, and binding them to real agent output is the first goal of Iteration 2.

The floor-distance CLI prototype demonstrated trained role detection, tracking, provisional calibration and conservative metric export; it was descoped because the moving-camera footage supported almost no floor projections, and body-height-normalised distance replaced it. The interaction pipeline now combines trained roles, proximity, duration and a pose judge.

Chapter 7

Testing and Early Validation

7.1 Testing Strategy

Testing combines functional checks of implemented modules, integration checks across the proxy, backend, and database, security checks of access control, and, in FYP-2, quantitative evaluation of the vision agents against manually labelled clips.

7.2 Test Levels and Techniques

- **Unit level:** role/classification, engagement, pass-by, pose-judge, tracking and benchmark-metric checks with known synthetic inputs; the focused floor-geometry, failure-state, smoothing, motion and evaluation checks belonged to the descoped floor prototype.
- **Vision CLI integration:** actual trained weights and local video, exported video readability, JSONL invariants and explicit unavailable-state checks.
- **Integration level:** frontend to proxy to FastAPI to SQLite for account/application flows.
- **System level:** end-to-end use of dashboards, reports and person-analysis outputs.
- **Security testing:** direct calls without cookie or secret.
- **Vision evaluation:** detector KPIs on a held-out split and interaction precision/recall against labelled clips through `backend/interaction/benchmark.py`; broader labelled evaluation continues in FYP-2.

7.3 Test Cases

Table 7.1: Test case summary

ID	Test	Expected result	Req.
TC-01	Log in as Manager using a valid existing account	Session cookie set; manager pages open	FR-10
TC-02	Open a protected manager application route without a cookie, or call backend without the secret	Request rejected	NFR-07/08
TC-07	Person detection on a labelled clip	Detection precision/recall reported	FR-01/02
TC-08	Staff/customer classification on a labelled clip	Accuracy reported	FR-03
TC-09	Controlled clip with unattended customer	Unattended flag and alert raised	FR-04/05
TC-10	Path and dwell on a labelled clip	Zone dwell matches annotation within tolerance	FR-07
TC-12	Navigate all pages for both roles	Pages render; employee blocked from manager pages	FR-11–13, 18–20
TC-13	Build report and export PDF	PDF contains selected sections and narrative	FR-17
TC-14	Uncalibrated trained-model preview	Role labels, IDs and foot proxies exported without fabricated metres	FR-21
TC-15	Headless landmark calibration	Provisional H, grid and floor map generated	FR-22/26
TC-16	Interactive point selection	Ordered clicks, undo/reset/save preserve original pixels	FR-22
TC-17	Known geometry and preprocessing	Expected projections, metre units and coordinate recovery	FR-23; NFR-11
TC-18	Invalid/unsupported positions	Explicit unavailable states rather than zero distances	FR-24; NFR-11

ID	Test	Expected result	Req.
TC-19	Camera alignment / no features	Shifted and unverified frames suppress supported output	FR-25
TC-20	Demo and diagnostic exports	Readable videos, timestamps and approximate/provisional labels	FR-27/28
TC-21	Pose judge decision logic	Facing heads confirm; heads turned away or backs to camera reject; no pose or a failing judge keeps the proximity result as `unverified`	FR-04
TC-22	Pass-by and moving-camera cases	A customer walking past produces no interaction; a pair under a panning camera still interacts	FR-04
TC-23	Benchmark metric functions	One-to-one temporal-IoU matching; correct precision, recall and F1	FR-03/04
TC-24	Interaction CLI on the local clip set	Every clip completes; annotated video, confirmed and rejected interactions written	FR-04; NFR-02
TC-25	Repeat run with the same inputs	Byte-identical JSON outputs	NFR-01

Detailed steps are in Appendix C.

7.4 Test Results

Table 7.2: Test execution status at mid-report

Test	Result	Notes
TC-01	Passed	Actual manager/employee account login, cookie flags, logout and wrong-credential rejection checked.
TC-02	Passed, adapted	Current settings proxy: anonymous 401, missing/wrong backend secret 401, employee access 403.
TC-13	Unavailable	Report/PDF route absent on current branch; corresponding route returned 404.
TC-12	Partial	Representative manager/employee views captured. Exhaustive role navigation not completed.
TC-07, TC-08	Measured	Held-out split (455 images): precision 0.963, recall 0.953, mAP50 0.969 (Table 7.4).
TC-09, TC-10	Pending	Unattended flags and trajectory are not yet produced (Iteration 2).
TC-21–TC-23	Passed	7 pose-judge, 2 pass-by (within the 8-test engagement suite) and 3 benchmark-metric tests.
TC-24	Passed	11 clips: 19 candidates, 17 confirmed, 2 rejected (Table 7.6).
TC-25	Passed	IMG_8829 JSON outputs byte-identical across two runs.

The application cases above were executed on 20 October 2026 against isolated SQLite databases and local development servers. All 15 retained in-scope HTTP assertions passed, including one assertion that the removed report route returned 404; that assertion is an absence check, not a feature pass. The role, engagement, identity and tracker suites were rerun for the corrected mid-iteration scope: 24 tests passed in 2.66 seconds. After the pose judge and benchmark were added, all interaction suites were rerun with pytest: 44 tests passed in 1.72 seconds (engagement 8, role classifiers 4 and 5, identity 4, tracker 5, pose judge 7, benchmark 3, frame processor 8). UI screenshots substantiate representative views only. Three warm login HTTP requests took 69.86, 77.12 and 82.83 ms (median 77.12 ms); these exclude compilation, rendering and inference and do

not establish a dashboard performance requirement.

Version boundary. Commit `8c3a6fe` removed the calibrated floor-distance source on 20 October. The floor results below preserve the earlier implementation evidence; they are not a successful rerun of removed source at current HEAD. The floor-distance software/integration evidence below was executed during the 19–20 October 2026 implementation session. The results below describe software checks, integration behavior and accepted-position coverage.

7.4.1 Floor-Distance Software and Integration Results

Table 7.3: Executed floor-distance checks and temporary-video coverage

Check / quantity	Actual result
Focused floor-distance automated suite (TC-17–TC-19 and evaluation functions)	17 tests passed; synthetic geometry/failure cases, smoothing/prediction, alignment shifts and error statistics.
Actual trained-model preview (TC-14)	Completed with local `models/best.pt`; no retraining.
Headless calibration (TC-15)	Completed; grid and versioned provisional calibration saved.
Point-click calibration (TC-16)	Point-selection interface implemented.
Sampled clip execution (TC-20)	148 processed frames, sampling every 30 source frames across the clip.
Detection observations	284 repeated observations, not 284 unique people.
Frame alignment outcomes	1 reference snapshot; 9 motion-invalid frames; 138 unverified frames.
Accepted position observations	2 of 284, approximately 0.70% accepted coverage.
Available approximate pair observations	1 reference-frame pair, including a seated-person proxy.
Video/record verification	Both 148-frame videos decode through the final frame; timestamps increase; unsupported positions have null coordinates.

Check / quantity	Actual result
Snapshot / missing-track behavior	Three-frame check preserves explicit unavailable selected-pair records.
Missing-weight handling	Actionable local missing-model error verified.

The executed commands are documented in `docs/floor_distance_usage.md`. The coverage summary is exported as `data/outputs/distance/demo/summary.json`, with observation records in `metrics.jsonl`. The 0.70% value is accepted-position coverage, not accuracy. The reference pair is an approximate box-proxy demonstration involving a seated person. No whole-clip supported metric result is claimed.

7.4.2 Interaction Pipeline and Benchmark Results

The benchmark (`backend/interaction/benchmark.py`) was run on 20 October 2026 on a CPU-only laptop, scoring and tracking every tenth frame; the full report is `evidence/benchmark_20261020.json`.

Table 7.4: Detector KPIs on the recording-level held-out split (455 images)

Class	Precision	Recall	mAP50	mAP50-95
All	0.963	0.953	0.969	0.896
Employee	0.968	0.990	0.982	–
Customer	0.958	0.916	0.955	–

Table 7.5: Interaction accuracy on the labelled clip (IMG_8829)

Decision	Precision	Recall	F1	TP / FP / FN
Proximity alone	0.50	1.00	0.67	1 / 1 / 0
Proximity + pose judge	1.00	1.00	1.00	1 / 0 / 0

The false positive removed by the judge is the 1.7–15.3 s segment in which the employee and customer walk away side by side (Figure 6.5, right). One clip with one true interaction shows the judge's effect; it does not establish accuracy. IMG_8829 is in the detector's training split, so it is used for interaction KPIs only.

Table 7.6: Interaction CLI on 11 local clips (scored every 10th frame, tracked every 3rd)

Clip	Candidates	Confirmed	Rejected	Duration (s)	Realtime
IMG_8829	2	1	1	8.0	1.40×
IMG_8878	2	2	0	15.7	1.54×
IMG_8880	1	1	0	25.3	1.41×
IMG_8881	3	3	0	38.0	1.31×
IMG_8885	0	0	0	0	1.37×
IMG_8886	1	1	0	6.7	1.23×
IMG_8890	2	2	0	31.7	1.40×
IMG_8893	3	2	1	37.0	1.43×
IMG_8894	1	1	0	26.7	1.16×
IMG_8895	3	3	0	19.7	0.80×
IMG_8897	1	1	0	23.0	1.23×
Total	19	17	2	231.8	

Candidates, confirmations and durations are from the CLI run; the realtime factor is from the benchmark run, which also tracks every tenth frame. Ten of the 11 clips are not labelled, so their counts are pipeline outputs, not accuracy; their candidates are listed in `evidence/interaction_judge_batch_20261020.csv` for labelling. No candidate was `unverified`. Per-frame latency was 120–171 ms at the median, and judge calls took 226–401 ms each, except on IMG_8895, where they took about 893 ms; that clip missed the benchmark's 1.0× realtime target and should be re-measured. With tracking on every third frame, processing ran at 0.50–0.59× realtime.

7.5 Acceptance Testing / Validation

Acceptance follows the criteria in Chapter 4. We showed the supervisor the 70-video dataset, which he agreed to; this was the material he requested. The dataset was approved at the supervisor meeting on 12 October 2026. The separate client meeting was on 10 September 2026; see Appendix F. No numeric accuracy or response-time targets were agreed. This is dataset approval rather than acceptance of measured classification, interaction or dashboard performance. Executed local checks and screenshots provide the separate software evidence.

7.6 Edge and Failure Cases

- Staff outside their usual zone or customers in uniform-like clothing, affecting classification.
- Identity switches during occlusion, affecting dwell and interaction durations.
- Expired or missing session cookie.
- Depth ambiguity in the interaction pipeline: people at different depths whose boxes overlap can appear close, and a head turned only slightly can pass the 60° facing test (IMG_8878 at 36 s).
- Identity switches on a moving camera split one customer into several track IDs and one interaction into several (IMG_8881, IMG_8895).
- Duplicate/collinear landmarks, invalid homogeneous denominators, incompatible units or frame sizes, stale homography and extrapolation outside calibrated coverage.
- Lower-frame truncation, manual exclusions, person overlap and seated/occluded feet; box-bottom positions remain proxies.
- Missing selected tracks, predicted positions, implausible metric jumps and expired reliable observations.
- Camera shifts, insufficient floor matches or poor spatial spread: preserve unavailable measurements rather than reuse stale alignment.

7.7 Defects and Corrective Actions

The first actual tracking run exposed a missing `lap` assignment dependency; it was installed into the existing virtual environment and declared in backend requirements, after which ByteTrack inference completed. Reference-frame selection was changed to use media timestamps rather than assuming constant-rate timing. A stale-calibration guard and explicit illustrative-config rejection prevent accidental use of incompatible inputs. During the current application checks, the development frontend exhausted a 768 MB Node heap; it was restarted with a 2048 MB heap, without modifying application source. The interaction CLI originally appended an active interaction once per scored frame; it now keeps one record per interaction. Annotated videos written while skipping frames keep the source frame rate and play faster than real time; review copies were re-encoded

at the true rate. Exhaustive page navigation was not completed. The first unit-test attempt encountered SQLite permission errors in the sandbox temporary directory; using a writable repository-local temporary directory resolved the temporary-directory issue; the corrected-scope rerun subsequently passed 24 tests.

7.8 Current Validation Status

The floor-distance prototype passed its focused synthetic software checks and actual-model/video integration checks before it was descoped. The moving temporary clip supported only the calibration-reference snapshot; the remaining sampled frames were motion-invalid or unverified. The exported quality states distinguish supported approximate positions from unavailable measurements. Trained roles, body-height proximity, duration and the pose judge are now combined in the interaction pipeline, which runs on all 11 local clips; its accuracy is measured on one labelled clip only, and dashboard integration is pending.

Chapter 8

Project Management and Development Evidence

8.1 Work Completed

- Next.js application with manager, employee, and shared pages.
- Role-aware access with signed session cookies and an internal secret header.
- Trained-role/ByteTrack foot-point preview and provisional floor-distance CLI: editable calibration, motion/reliability checks, metric smoothing and diagnostic exports (since descoped).
- Two-stage interaction pipeline: trained role detector, BoT-SORT tracking, body-height proximity with a pass-by filter, and a YOLO pose judge; plus a benchmark for detector, interaction and speed KPIs.
- Executed 17 historical focused floor-distance checks; 44 current interaction unit tests and 5 handoff tests pass; detector benchmarked on 455 held-out images and the interaction pipeline run on 11 local clips.

8.2 Work Remaining

Integrating the vision agents with the dashboard, labelled detector/interaction evaluation, the Insight Narration agent on real data, and hardening. Details are in Chapter 9.

8.3 Individual Responsibilities

Table 8.1: Responsibilities

Member	Responsibility
Moiz Ansari (23I-0523)	Frontend; distance between people; head, hips and feet band handling when the full-person position is rejected
Saqib Atif (23I-0769)	Customer/employee pose detection and testing
Ruhab Ahmad (23I-0559)	Frontend, report and classification

8.4 Contribution of Each Team Member

Moiz worked on frontend development, person-to-person distance and the head/hips/feet band approach for rejected full-person positions; Saqib worked on customer/employee pose detection and testing; Ruhab worked on frontend development, the report and classification. The head/hips/feet categories describe body regions used for distance estimation, not employee/customer role classes.

Git records attribute the provisional floor-distance implementation (`4a9023a`) and report addition (`80fb3fd`) to Moiz Ansari. They attribute the interaction detector (`e3fe2af`) and video-verification record (`6c25090`) to Ruhab Ahmad. The Git author `msaqibatifj` is recorded for temporal role classification (`882f7dc`), the shared OpenVINO model (`e2b71a5`) and moving-camera engagement (`face2b5`). Git author names are shown as recorded. These are commit-author records, not a claim that each author exclusively performed every change or a complete account of non-code work.

8.5 Repository and Version Control Evidence

The project is kept in the Git repository `muhammad-moiz-ansari/retail-store-analysis` on GitHub. Work is organised on feature branches, and this report corresponds to the `feature/interaction` branch, inspected on 20 October 2026 at commit `6560403`.

8.6 Milestones and Commit Evidence

The actual floor-distance implementation commit is `4a9023a` (19 October 2026, Moiz Ansari), titled “feat(vision): add provisional floor-distance estimation pipeline”. A later

commit removed the calibrated floor-distance pipeline (`8c3a6fe`, 20 October), attributed to Git author `msaqibatifj`. Therefore historical implementation evidence must be read with its commit context. The on-screen interaction flag and one-record-per-interaction output were committed on `feature/interaction-flag` (`20db4c6`, `ee7947f`, `0411408`, 19 October 2026, Git author `msaqibatifj`); the pose judge and benchmark were developed there and ported to `feature/interaction` in commit `6560403` (20 October 2026), after unused code was removed in `cffa30b`. No Git tags were present in the inspected repository. Selected commit and author-count evidence is reproduced in Appendix D; no project-board or supervisor milestone evidence is inferred from commit dates.

8.7 Tools and Tool-Assisted Development

Development tools include Node.js and npm, a Python virtual environment, Git and GitHub, and the libraries listed in Chapter 6. We also used generative-AI assistants, disclosed in Table 8.2.

Table 8.2: Generative-AI tool use and verification

Tool	Used for	How the output was checked
Codex	Floor-distance implementation (since descoped), diagnostic runs, and revision of the report source.	Focused automated tests, local trained-model/video runs, inspection of preview and grid outputs, export readability and record-invariant checks.
Claude Code	Porting the pose judge and benchmark to the main branch, the batch and benchmark runs, and the 20 October report revision.	Automated tests, a byte-identical comparison of pipeline outputs before and after the port, and inspection of annotated frames.
Claude	General AI/development assistant.	Reviewed by the team member who owns the affected module (Table 8.1).

AI assistance does not replace the recorded test and review evidence, and each team member must be able to explain the code and text in the area they own.

8.8 External Components and Dependencies

Table 8.3: External components

Component	Use
YOLO implementation [7]	YOLOv8n pose model as interaction judge; standard YOLO detection for customer/employee classes
OpenVINO	CPU inference runtime for the role detector
ByteTrack [9] / BoT-SORT	Per-camera tracking (BoT-SORT with appearance re-identification in the interaction pipeline); `lap` supplies assignment support
OpenCV / NumPy [12]	Image processing and video input/output; homography fitting, projection and ORB matching in the descoped floor prototype
ORB [15] and RANSAC [14]	Reference-alignment evidence and robust fitting in the descoped floor prototype
FastAPI [16]	Application and camera API
Next.js and React [17]	Frontend
Recharts, Tailwind CSS	Charts, styling
SQLite	Application, job and interaction storage

Table 8.4: Dependency licence review from installed metadata and upstream notices, 20 October 2026

Component	Licence / scope
Ultralytics implementation	AGPL-3.0 or a separate Enterprise licence, according to its publisher [18]. Academic use does not by itself remove applicable obligations.
Original ByteTrack repository	MIT [19]; this does not replace the licence of the Ultralytics implementation used by the project.
OpenCV	Apache-2.0 for current releases [20].
OpenVINO	Apache-2.0 (installed metadata, version 2026.4.1).
FastAPI, pytest	MIT (installed distribution metadata).
Uvicorn, HTTPX; lap	BSD-3-Clause; BSD-2-Clause for lap (installed metadata).

Component	Licence / scope
NumPy and PyTorch	BSD-family and bundled-component notices; retain the complete installed distribution notices rather than assuming a single licence covers every component.
Pillow	MIT-CMU (installed metadata).
Next.js, React, ReactDOM, Recharts, Tailwind CSS	MIT (installed package metadata).
Inter, Space Grotesk, JetBrains Mono	SIL Open Font License 1.1 [21, 22, 23].
SQLite	Public-domain software; the application's data rights remain separate.

This is a dependency-notice inventory, not a certification of licence compliance or ownership of trained weights/video. Distribution should preserve applicable notices and satisfy the relevant licence terms. Versions installed during testing can differ from manifest ranges; the test environment contained Next.js 16.3.3, React 19.2.8 and Ultralytics 8.4.138.

8.9 Risks and Mitigation

Table 8.5: Risk register

Risk	Level	Mitigation
CPU-only inference too slow	High	Frame sampling, small models, batch processing
Staff/customer misclassification	High	Zone rules plus appearance, labelled evaluation, manager review
Limited footage for evaluation	Medium	Record or label additional clips early in FYP-2
Integration slippage	Medium	Bind one agent at a time, keep prototype data fallback
Privacy or misuse concerns	Medium	Non-biometric design, decision-support framing
Depth ambiguity in a single image	High	Height-ratio check and pose judge; labelled evaluation; reported as a limitation
Identity switches on moving footage	High	BoT-SORT re-identification and relinking; count split interactions during evaluation
Pose judge too lenient or too strict	Medium	Benchmark with and without the judge; tune facing angle and fraction on labelled clips

8.10 Security and Privacy Considerations

Manager routes require a signed HttpOnly session cookie, and direct backend calls require an internal secret header that only the server-side proxy supplies. Secrets come from environment variables. The current login verifies stored account credentials; the earlier arbitrary-email/six-character demo rule no longer describes the tested implementation. The access-control checks are described in Chapter 7.

Floor footage, calibration frames/configuration and generated diagnostics remain local under Git-ignored data directories. The report includes only selected local demonstration frames; no footage upload, persistent person crops or facial-identity records are required by the prototype.

8.11 Current Limitations

- Vision-derived dashboard metrics still use prototype or demo data; system status is available as a backend endpoint only.
- The pipeline is offline/batch, not real time.
- Single-store prototype; no multi-store deployment is claimed.
- Current authentication uses stored accounts; role-navigation checks cover representative views rather than every application route.
- The floor-distance prototype was descoped, so no distances in metres are produced; interaction distance is relative to body height and cannot fully resolve depth.
- Interaction accuracy is measured on one labelled clip only; identity switches on moving footage can split one interaction into several.
- Interaction results are available through the CLI; dashboard integration remains pending.

Chapter 9

Remaining FYP Plan

9.1 Remaining Scope

Integrate and evaluate the vision agents, bind real outputs to the dashboard, and prepare the final demonstration. Items listed as out of scope in Chapter 1 remain future work.

9.2 Remaining Modules and Features

- Integrate and evaluate Detection, Classification, Engagement and Trajectory agents; detection, classification and interaction detection with the pose judge are already implemented in the interaction CLI.
- Evaluate interaction detection (proximity, duration, pass-by filter and pose judge) on labelled clips, including close pass-bys that are not assistance; ten local clips await labels.
- Insight Narration agent over real analytics.
- Real data in Dashboard, Customer Info, Employee Info, Spatial Analytics, Alerts, and System Status.
- Implement camera handoff in the next phase: associate a person leaving one camera with a person entering another using timing, direction and appearance evidence. This is outside the mid-iteration deliverables; ambiguous associations must remain unresolved. An experimental two-camera handoff script and `/camera/*` job routes already exist.

9.3 Remaining Iterations

Table 9.1: Planned FYP-2 iterations

Iteration	Focus
2	Vision-component integration, real data in dashboard pages and planned cross-camera association
3	Evaluation on labelled clips, tuning of thresholds and heuristics, alerts on real events
4	Hardening, security, performance, documentation, and final demonstration

9.4 Implementation Plan

Bind agents one at a time to the dashboard, beginning with Detection and footfall, then Classification, Engagement, and Trajectory, keeping prototype data as a fallback until each agent passes its checks. Develop and evaluate cross-camera association separately from same-camera interaction detection.

9.5 Testing and Validation Plan

Label a set of clips for staff/customer, interactions, and zone dwell. Report precision, recall, and error analysis per agent. Re-run Appendix test cases after each integration and record results. Check performance on CPU and record throughput.

9.6 Deployment Plan

The deliverable is a local deployment run with the documented `npm run dev:all` flow and production build commands (`npm run build`, `npm start`). A pilot with a real retail partner is future work.

9.7 Tentative Timeline

Table 9.2: Tentative FYP-2 timeline (to be confirmed with the supervisor)

Period	Activity
Weeks 1–3	Detection and footfall integration
Weeks 4–6	Classification and engagement integration
Weeks 7–9	Trajectory, narration, dashboard binding
Weeks 10–12	Labelled role/interaction evaluation and tuning
Weeks 13–14	Hardening, security, documentation
Weeks 15–16	Final testing, demonstration, and report

9.8 Expected Final Deliverables

- Working RetailPulse system with real vision analytics and role-aware web access.
- Evaluation report with detector and interaction results (with and without the pose judge) and CPU throughput.
- Source repository with documentation and run instructions.
- Final report and demonstration.

Chapter 10

Conclusion

Retailers hold CCTV footage that could answer everyday operational questions but is rarely used for that purpose, and small and mid-sized stores often cannot afford specialised analytics. RetailPulse addresses this with a CPU-friendly, non-biometric, multi-agent design that reuses existing cameras. Research on detection, tracking, and re-identification, and a feasibility triage, shaped the scope: a single-store, three-camera prototype focused on person detection, role identification and pose-checked interaction detection.

At the mid-report stage the requirements and architecture are defined. The web application with manager and employee roles, secure session handling and person-analysis prototypes form the current implementation; report export remains a planned feature on the tested branch. An interaction pipeline combines the trained employee/customer detector, body-height-normalised proximity and a YOLO pose judge that confirms an interaction only when someone faces the other person; on the one labelled clip the judge raises interaction precision from 0.50 to 1.00, and the detector reaches mAP50 0.969 on 455 held-out images. A floor-distance CLI passed seventeen focused checks before it was descoped, and 44 current interaction unit tests pass. Dashboard vision metrics still rely on prototype data. The main risks are CPU throughput, classification accuracy, and interaction interpretation. FYP-2 will integrate the vision agents, evaluate them on labelled store footage, and harden the system for the final demonstration.

Appendix A

Proposal Action Register

The response register is in Chapter 3. The following comments are the proposal-defence feedback of 11 September 2026 as recorded by the team. No signed feedback form is attached.

A.1 Comments / Recommended Modifications

> Students have no clue what they are doing. Technical knowledge is lacking. The students do not have an understanding of the project. Students need to learn more about the requirements of the project and next time be well prepared. Should work on their presentation skills; the presenters kept pacing back and forth. Their project was poorly explained, there was no clarity on the exact features, and the overall idea was unclear. Also, they did not defend the generalizability of their project; the behavioral requirements of employees vary from retail to retail store (for example, luxury brands like Gucci, Prada, etc. don't want their employees to be constantly approaching customers), so how is their project universal for all types of retail stores? Improve your skills. It is a healthy project if it is properly focused.

A.2 Response Priorities

Improve technical preparation and feature explanations; focus the project; justify store-specific behavioural interpretation; rehearse a clearer presentation. The action register distinguishes report changes from work still planned, rather than treating the feedback as resolved solely by adding text.

Appendix B

Detailed Requirements

B.1 Requirement Priorities

High-priority requirements: FR-01 to FR-05, FR-07, FR-10 to FR-13, FR-29 and FR-30 to FR-32. Medium: FR-06, FR-17, FR-18, FR-33 and FR-34. Low: FR-19, FR-20. Security requirements NFR-07 to NFR-10 apply to all modules.

B.2 Additional Use Case: UC-11 View personal performance

Actor	Employee
Precondition	Employee is authenticated.
Main flow	1. Employee opens My Workspace and sees shift overview, today's score, zone assignment, and updates. 2. Employee opens My Performance and sees the rubric breakdown, trend charts, and daily log. 3. Employee selects a day to see its details.
Postcondition	Only the employee's own data is shown.

B.3 Floor-Distance Requirement Priorities and Operator Use Case

FR-21–FR-26 and FR-28 are high priority; FR-27 is medium priority. NFR-11 applies to all floor-distance paths. UC-14 in Chapter 4 covers the operator-facing calibration

and diagnostic workflow. It supplements existing manager/employee use cases and does not imply that a calibration interface or metric interaction feature is deployed in the web application. The prototype that implemented these requirements was descoped in commit `8c3a6fe` (Section 6.4.3).

Appendix C

Detailed Test Cases

Table C.1: Detailed test cases: application and interaction

ID	Steps	Expected	Result
TC-01	Log in with a valid existing Manager account; test wrong password, role mismatch and logout.	Signed cookie and dashboard; invalid credentials rejected.	Passed 20 Oct 2026
TC-02	Request current settings proxy anonymously; call backend with missing/wrong secret; try employee credentials.	HTTP 401/403 as appropriate.	Passed adapted case
TC-12	Inspect representative manager/employee views and route-guard source.	Role-specific pages and consistent restrictions.	Partial; representative views only; not exhaustive
TC-13	Build a report and export PDF.	PDF opens with selected sections.	Unavailable: report route removed (404)
TC-07, TC-08	Run the benchmark with the detector section enabled on the held-out split.	Overall and per-class precision, recall and mAP.	Measured 20 Oct 2026
TC-09, TC-10	Defined in FYP-2 against labelled clips.	See Chapter 7.	Pending

ID	Steps	Expected	Result
TC-21–TC-23	Run `test_judge.py`, `test_engagement.py` and `test_benchmark.py`.	All checks pass.	Passed
TC-24	Run the interaction CLI with `-every 10 -track-every 3` on each local clip; reopen videos and JSON.	All clips complete; confirmed and rejected interactions written.	Passed (11 clips)
TC-25	Run the CLI twice on IMG_8829 with identical settings; compare JSON files.	Byte-identical outputs.	Passed

C.1 Floor-Distance Test Procedures (Descoped Prototype)

Table C.2: Detailed test cases: descoped floor-distance prototype

ID	Steps	Expected outcome	Executed status
TC-14	Run uncalibrated preview on the local video with trained weights; inspect labels, IDs, red points and records.	Readable preview; no invented metres. Missing weights produce an actionable error.	Passed integration
TC-15	Fit the saved estimated A–B–D–C landmarks headlessly; inspect grid and generated JSON.	Correct units/order and provisional status; H and residuals saved.	Passed
TC-16	Omit headless mode; click numbered points, undo/reset, save and compare original-pixel coordinates.	Explicit correspondence order and correct display-coordinate recovery.	Interface implemented
TC-17	Run focused synthetic projection, distance, robust-fit, unit/frame and preprocessing tests.	Expected numerical geometry; reject incompatible/stale calibration.	Automated checks passed

ID	Steps	Expected outcome	Executed status
TC-18	Exercise invalid boxes/denominators, occlusion/exclusion, out-of-region, prediction and missing tracks.	Null/unavailable distances with explicit reasons; no fabricated zeroes.	Automated and CLI checks passed
TC-19	Apply synthetic camera shift and no-feature frames; run the temporary clip monitor.	Shifted/unverified frames suppress supported metric output.	Passed tested cases
TC-20	Run sampled demo; reopen both videos, inspect timestamps, provisional metadata and null coordinates.	148 frames in each video; valid records with explicit approximate/failure states.	Passed

The focused suite is invoked from the repository root with

```
.venv/Scripts/python.exe -m unittest
  backend.tests.test_floor_distance -v
```

It contains 17 tests. The local coverage evaluation command is

```
.venv/Scripts/python.exe scripts/evaluate_floor_distance.py
  -metrics data/outputs/distance/demo/metrics.jsonl
  -output data/outputs/distance/demo/evaluation.json
```

C.2 Current-Branch Reproduction

The 20 October check ran these existing files with pytest: `backend/interaction/test_engagement.py`, `test_cust_emp_classifier.py`, `test_role_classifier.py`, `test_identity.py`, `test_tracker.py` (all under the same interaction directory). Set `PYTHONPATH` to the backend directory and `TEMP`/`TMP` to a writable local test directory. Corrected-scope rerun: 24 passed in 2.66 seconds. After the pose judge and benchmark were added, `python -m pytest backend/interaction -q` passed 44 tests in 1.72 seconds. The benchmark is run with `python -m backend.interaction.benchmark backend/interaction/benchmarks/benchmark.json`. Historical floor-suite commands above refer to the implementation before its removal, not current-branch availability.

Appendix D

Repository and Contribution Evidence

Repository: `github.com/muhammad-moiz-ansari/retail-store-analysis`. Inspected branch: `feature/interaction`; HEAD: `6560403`; evidence date: 20 October 2026.

Table D.1: Selected commits verified from local Git history

Hash	Date	Git author	Recorded change
`e3fe2af`	16 Oct 2026	Ruhab Ahmad	Add interaction detection model.
`6c25090`	16 Oct 2026	Ruhab Ahmad	Record video interaction verification.
`882f7dc`	16 Oct 2026	`msaqibatifj`	Weighted customer/employee classifier with three-second hold.
`4a9023a`	19 Oct 2026	Moiz Ansari	Add provisional floor-distance estimation pipeline.
`e2b71a5`	19 Oct 2026	`msaqibatifj`	Single OpenVINO role model.
`face2b5`	20 Oct 2026	`msaqibatifj`	Distance-based engagement for moving camera.
`8c3a6fe`	20 Oct 2026	`msaqibatifj`	Remove calibrated floor-distance pipeline.
`80fb3fd`	20 Oct 2026	Moiz Ansari	Add mid-1 report.
`20db4c6`	19 Oct 2026	`msaqibatifj`	On-screen interaction flag (`feature/interaction-flag`).
`4cbe655`	20 Oct 2026	`msaqibatifj`	Remove unused vision-language-model stage.
`cffa30b`	20 Oct 2026	`msaqibatifj`	Remove unused code (interaction dataset, evaluation and model loader).

Hash	Date	Git author	Recorded change
6560403	20 Oct 2026	msaqibatifj	Pose judge, pass-by filter and benchmark.

D.1 Authorship and Milestone Evidence

At the inspected HEAD, `git shortlog -sne HEAD` records 43 commits for Moiz Ansari, 41 for `msaqibatifj` and 28 for Ruhab Ahmad, including merge commits. Counts describe recorded authorship and are not measures of effort or contribution quality. The mapping of account aliases and non-code responsibilities requires team confirmation. `git tag -list` returned no tags. No task-board history or supervisor-approved milestone record was available for this check, so none is fabricated. Textual Git evidence is supplied instead of an unverified contribution-graph screenshot.

D.2 Floor-Distance Implementation Evidence

Commit `4a9023a` records the shared inference helper, geometry/calibration/motion/smoothing/evaluation modules, four CLI scripts, focused tests, camera schema/examples and usage documentation. The selected report figures preserve local prototype outputs from that implementation. Commit `8c3a6fe` subsequently removed the source pipeline; current-branch availability must not be inferred from these historical figures. Raw video and generated diagnostics remain outside the report's tracked evidence. The new test summary is at `evidence/verification_20261020.json`; the benchmark report and interaction batch results are at `evidence/benchmark_20261020.json` and `evidence/interaction_judge_batch_20261020.csv`.

Appendix E

Additional Design Diagrams

The main architecture, tier and domain diagrams are in Chapter 5 (including Figures 5.1 and 5.2). Figure E.1 is derived from the current SQLite foreign-key relationships in `backend/store_core.py`; it shows a selected subset, not the complete conceptual domain model.

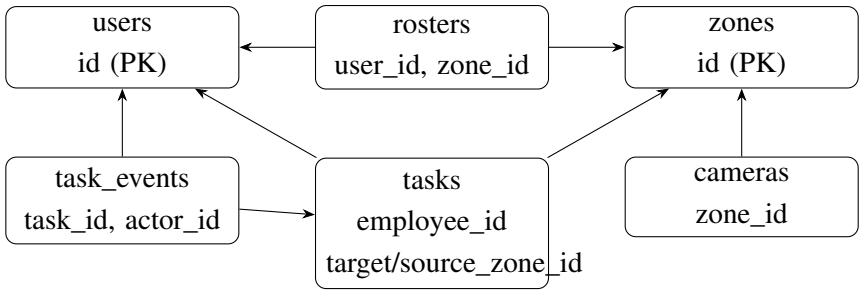


Figure E.1: Selected implemented SQLite relationships; arrows point from referencing tables to referenced tables

E.1 Floor Calibration Schema and Failure States

The floor process is shown in Figure 5.5; actual preview, calibration and map examples are in Figures 6.6–6.8. The authoritative correspondence order for the provisional rectangle is A–B–D–C. Floor coordinates are $(0,0)$, $(w,0)$, (w,d) and $(0,d)$ in metres for that order.

Record group	Key information
Identity / geometry	Schema version, camera/calibration ID, source reference, frame index/timestamp, frame size, coordinate space.
Landmarks / mapping	Original-pixel and metre correspondences, authoritative dimensions, image-to-floor H, inliers/residuals.
Operational quality	Supported polygon, exclusions, distortion handling, motion thresholds, smoothing/prediction limits.
Measurement status	Estimated calibration inputs and approximate-output metadata.
Runtime observation	Track ID, role/score, pixel foot proxy, nullable floor coordinates, motion/position state and reasons.

States include `uncalibrated`, `observed_proxy`, `predicted`, `unreliable`, `out_of_region`, `motion_unverified` and `motion_invalid`. Only supported observed proxies contribute to approximate pair distances. The localization equations are given in Equations 6.1–6.3; no 3D reconstruction is implied. These records belong to the descoped floor prototype (Section 6.4.3).

E.2 Interaction Verdict States

Each interaction candidate in the current pipeline carries one verdict; annotated output is in Figure 6.5.

Verdict	Meaning
`interaction`	The pose judge found a head turned towards the other person in at least half of the judged frames.
`no_interaction`	Poses were found but nobody faced the other person; written to `.rejected_interactions.json` and re-judged every four seconds while the pair stays close.
`unverified`	Judge switched off, no pose found, or the judge failed; the proximity result stands.

Appendix F

Meeting and Progress Evidence

Supervisor: Mr. Shehreyar Rashid. Dates, discussions and outcomes below are recorded from the team's recollection of each meeting.

Table F.1: Meeting and progress record

Date	Discussion	Outcome / Action
19 August 2026	Lack of a dataset; retailers' reluctance to share footage because of privacy concerns; the proposed project approach.	Supervisor suggested ways to address dataset access and recommended creating the team's own dataset for the time being.
8 September 2026	Review of the draft proposal-defence slides.	Supervisor advised additions and corrections and set a deadline to finalise and show the slides before the proposal defence. The deadline date was not recorded.
10 September 2026	Client meeting (Thursday).	Discussion details were not recorded.
5 October 2026	Dataset progress and presentation schedule.	Supervisor set a deadline for showing the dataset. The deadline date was not recorded.
12 October 2026	Additional supervisor meeting; presentation of the 70-video dataset.	Supervisor accepted the 70-video dataset.

Date	Discussion	Outcome / Action
14 October 2026	Work-progress update and the mid-evaluation deadline.	Supervisor requested a work update and communicated the mid-evaluation deadline; the recorded mid evaluation is 21 October 2026.

Dataset agreement. The supervisor accepted the 70-video dataset shown by the team. No numeric classification accuracy, interaction-detection accuracy or dashboard response-time threshold was agreed in that discussion. Dataset acceptance does not constitute approval of measured model performance.

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